



## **Module 5: Motor and naturalistic behavior**

### Tutorial 1: From tracking data to latent dynamics

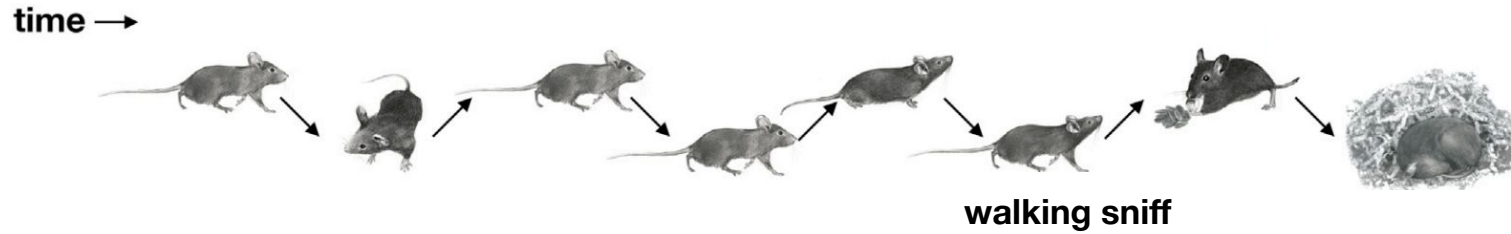
BAMB! 2026  
Heike Stein

Natural behavior is organized  
by time

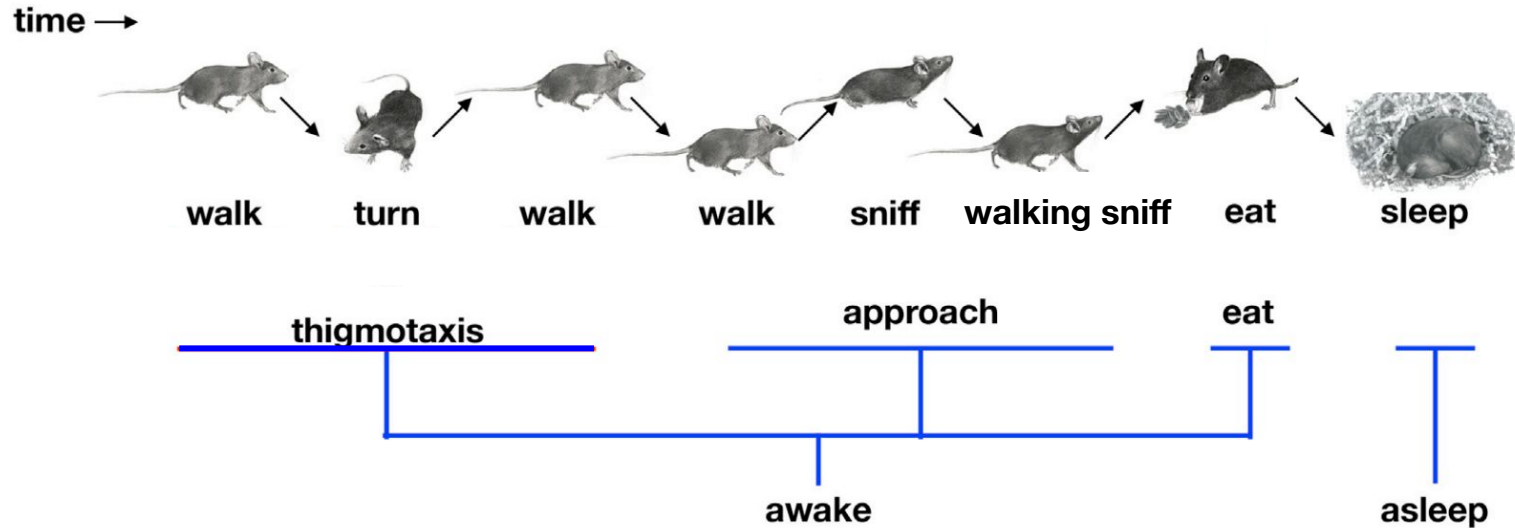




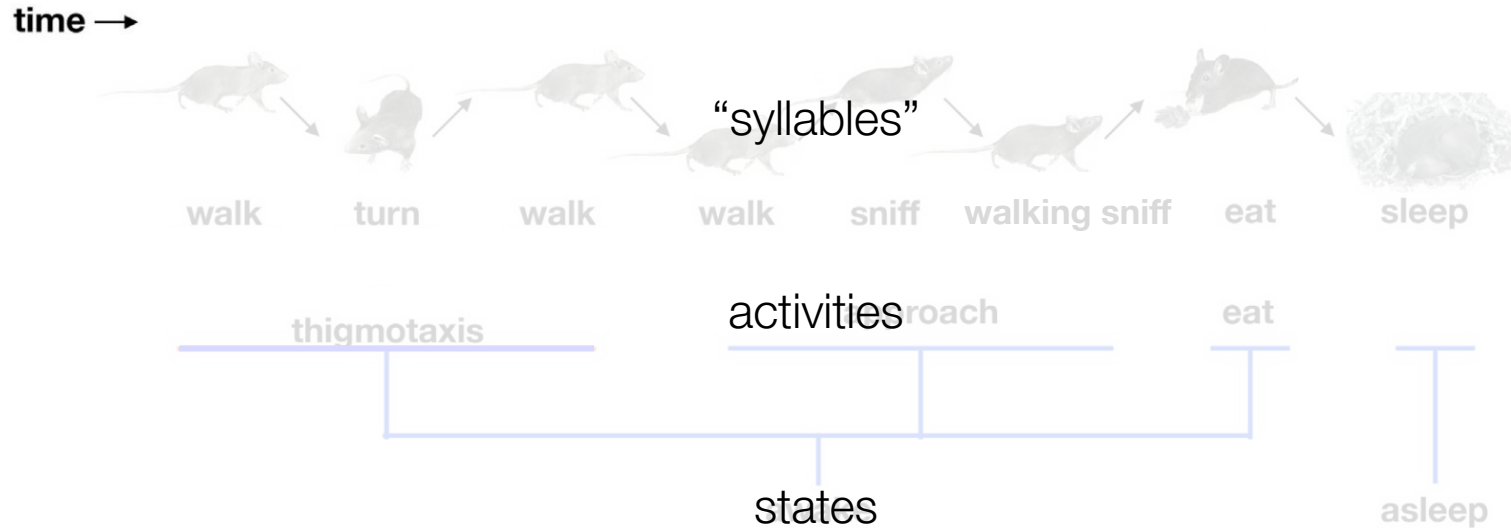
# Natural behavior is hierarchical



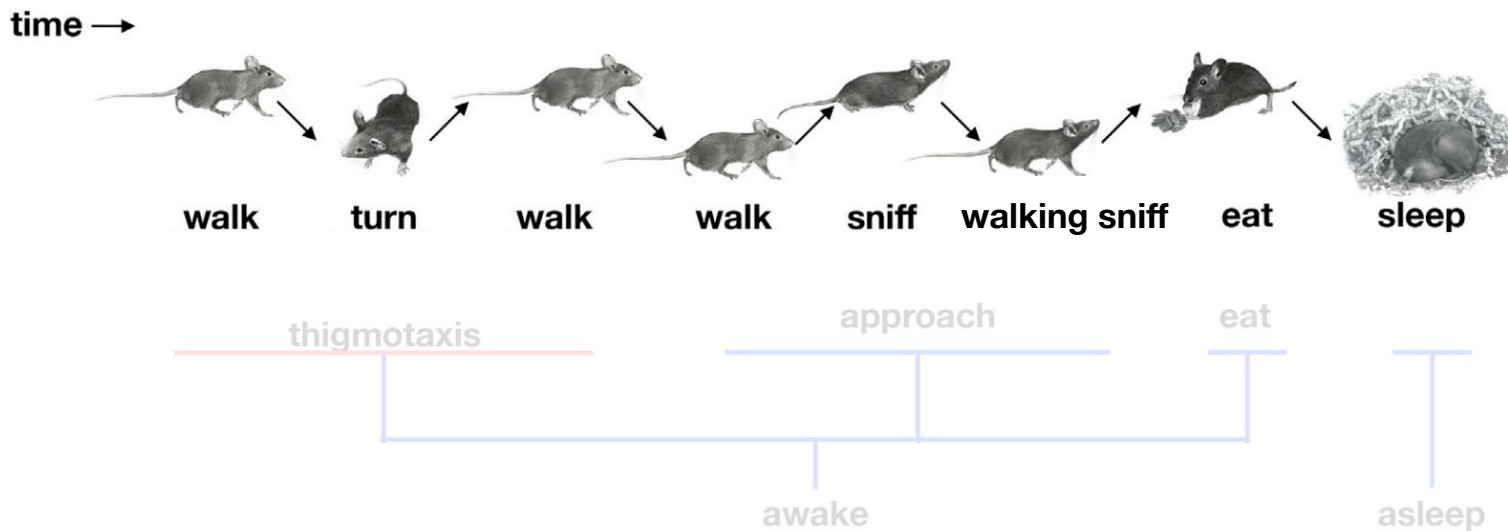
# Natural behavior is hierarchical



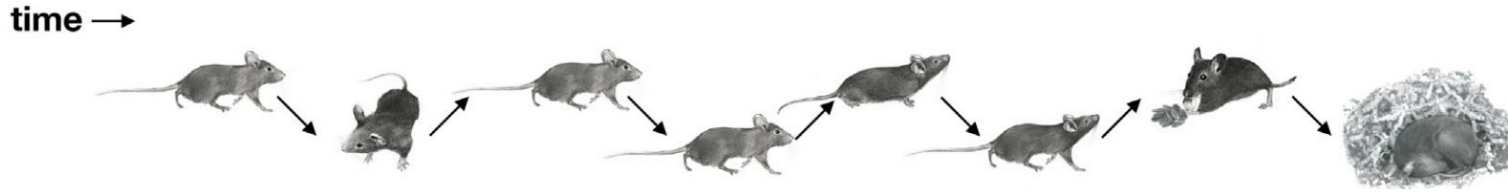
# Natural behavior is hierarchical



# How can we extract behavioral syllables?



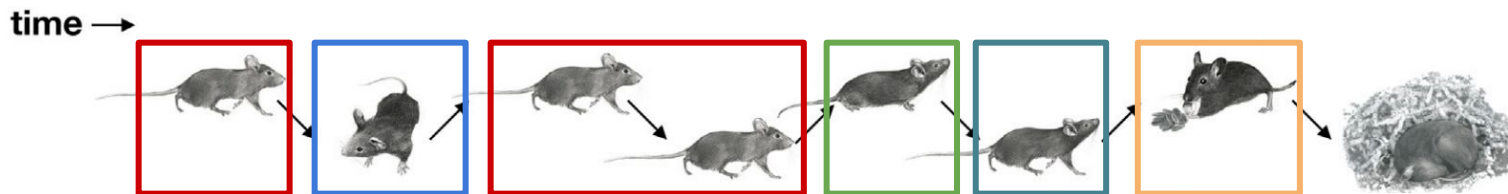
# How can we extract behavioral syllables?



**Idea: Characteristic movement within each syllable is described by simple, low-D dynamics**

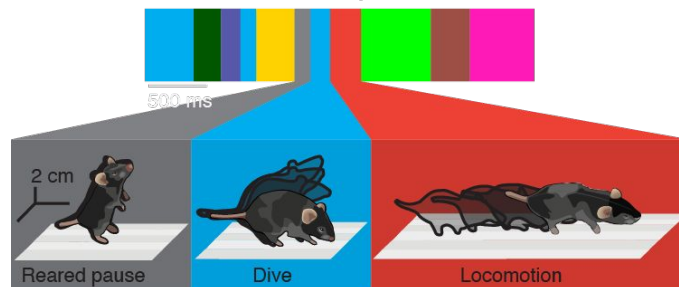


# How can we extract behavioral syllables?

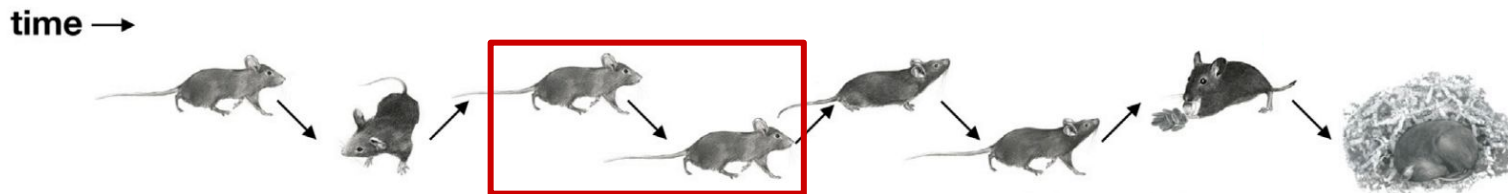


**Idea: Characteristic movement within each syllable is described by simple, low-D dynamics**

1. Identify syllable structure from time series

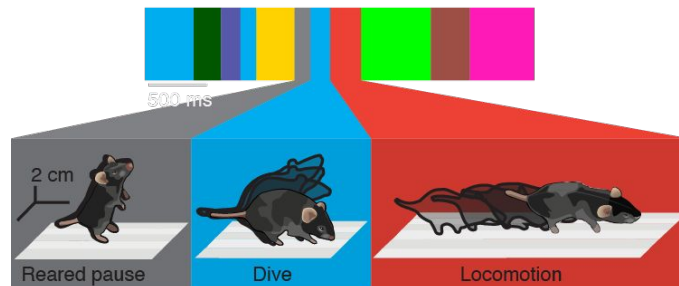


# How can we extract behavioral syllables?

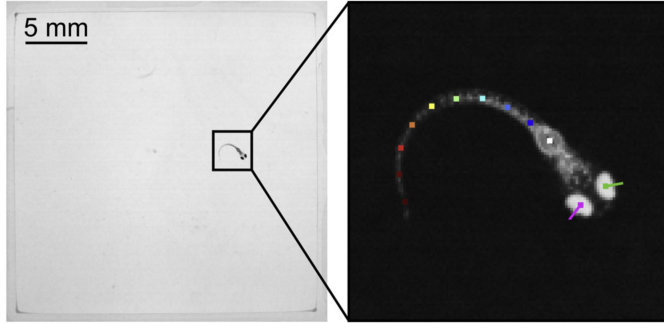


**Idea: Characteristic movement within each syllable is described by simple, low-D dynamics**

1. Identify syllable structure from time series
2. Model dynamics within each syllable

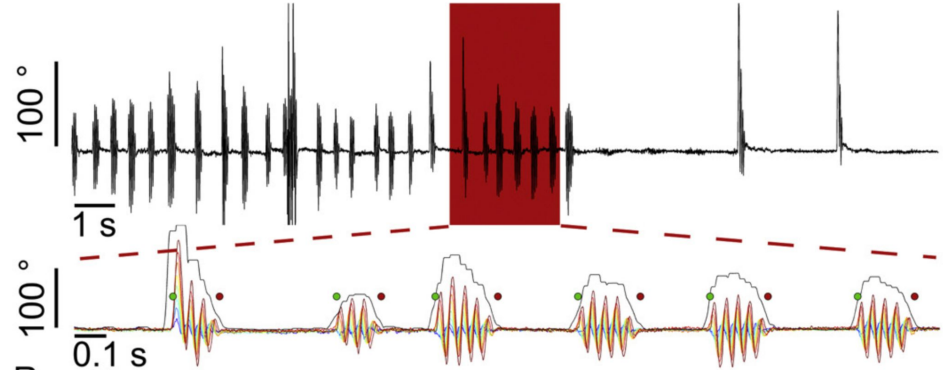
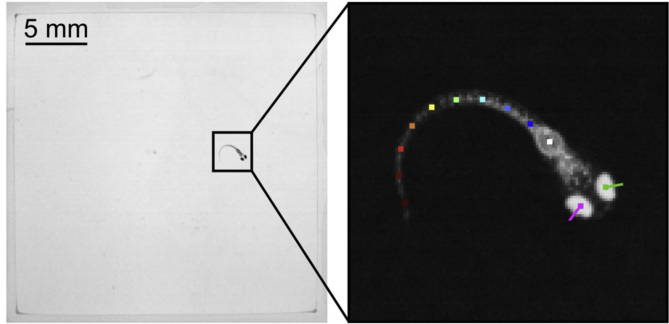


# Zebrafish locomotion: Bouts and tail oscillations



# Zebrafish locomotion: Bouts and tail oscillations

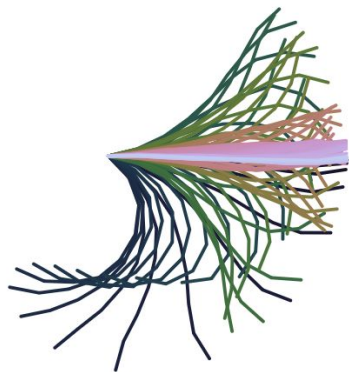
**Code: Part 0**



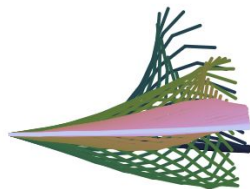


# Aligned and recentered body positions

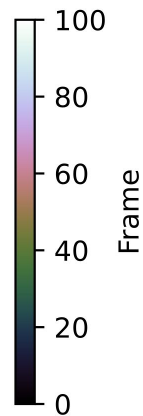
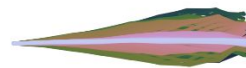
4: O-bend



8: Routine Turn

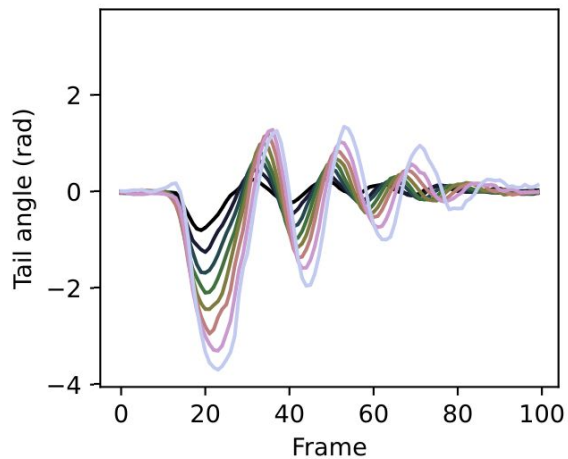


11: Approach Swim

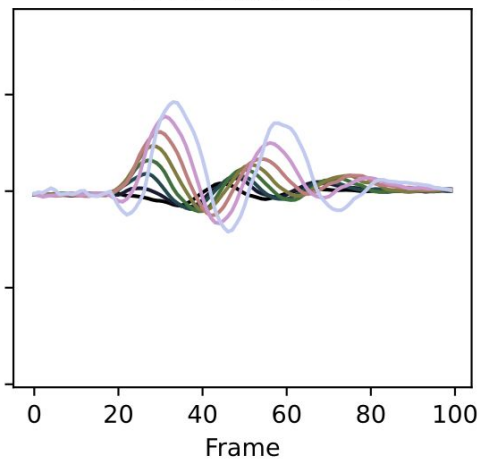


# High-dimensional timeseries

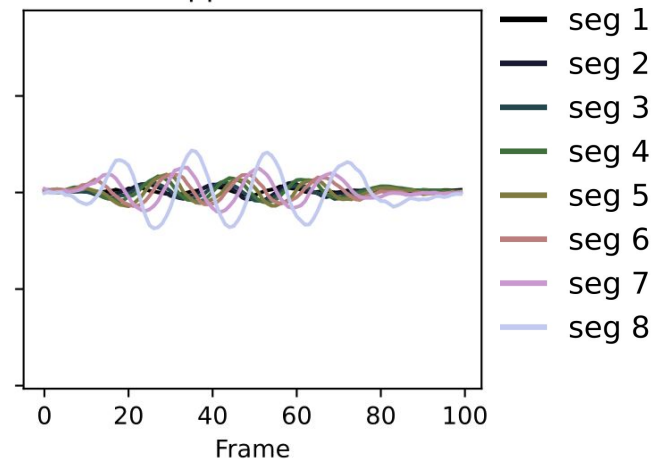
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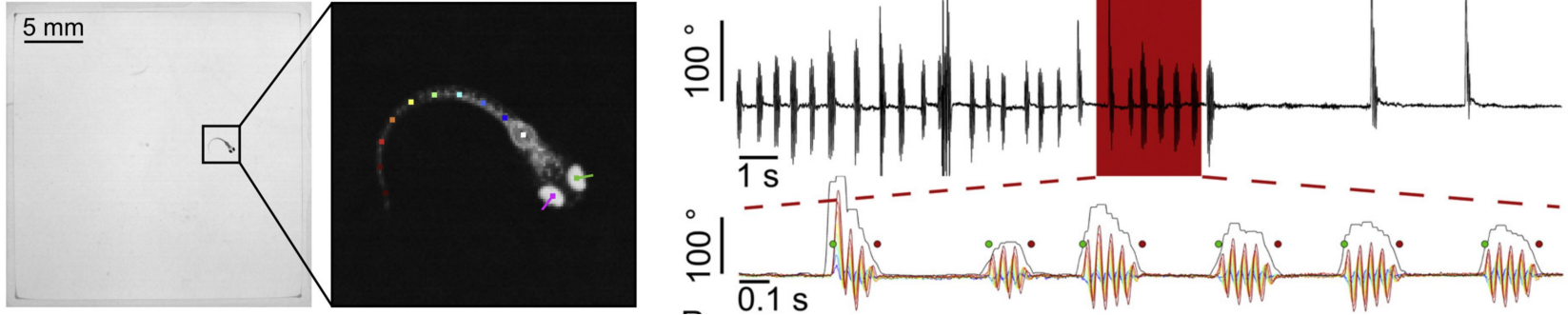


- seg 1
- seg 2
- seg 3
- seg 4
- seg 5
- seg 6
- seg 7
- seg 8





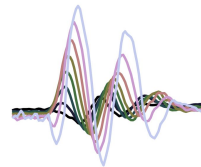
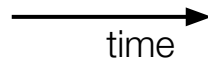
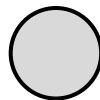
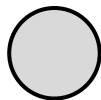
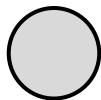
# Zebrafish locomotion: Bouts and tail oscillations



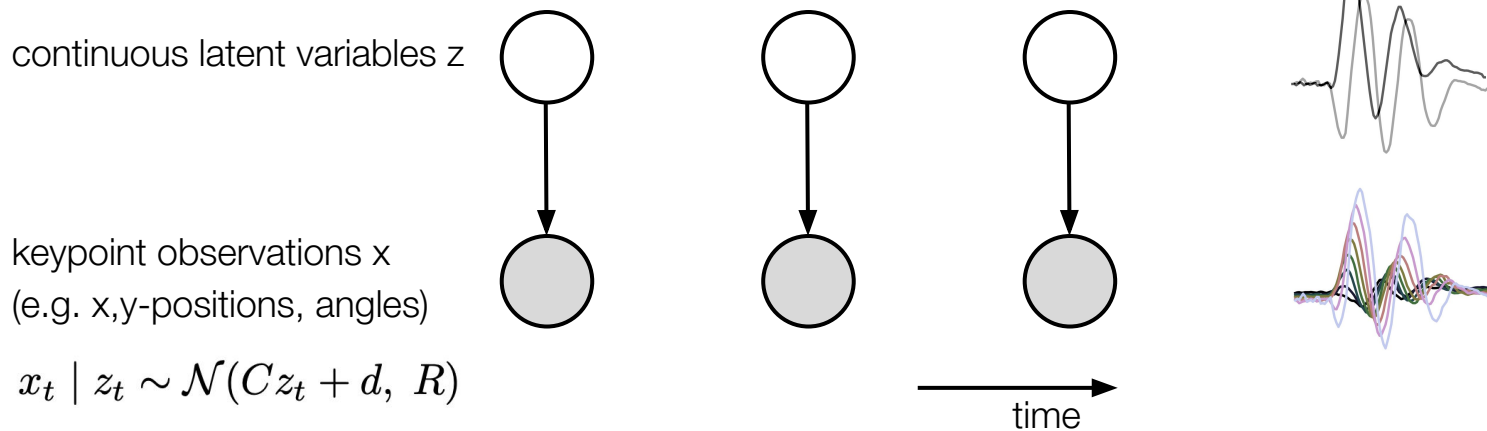
1. Identify syllable structure from time series
2. Model dynamics within each syllable

# Modeling motor time series with hierarchical structure

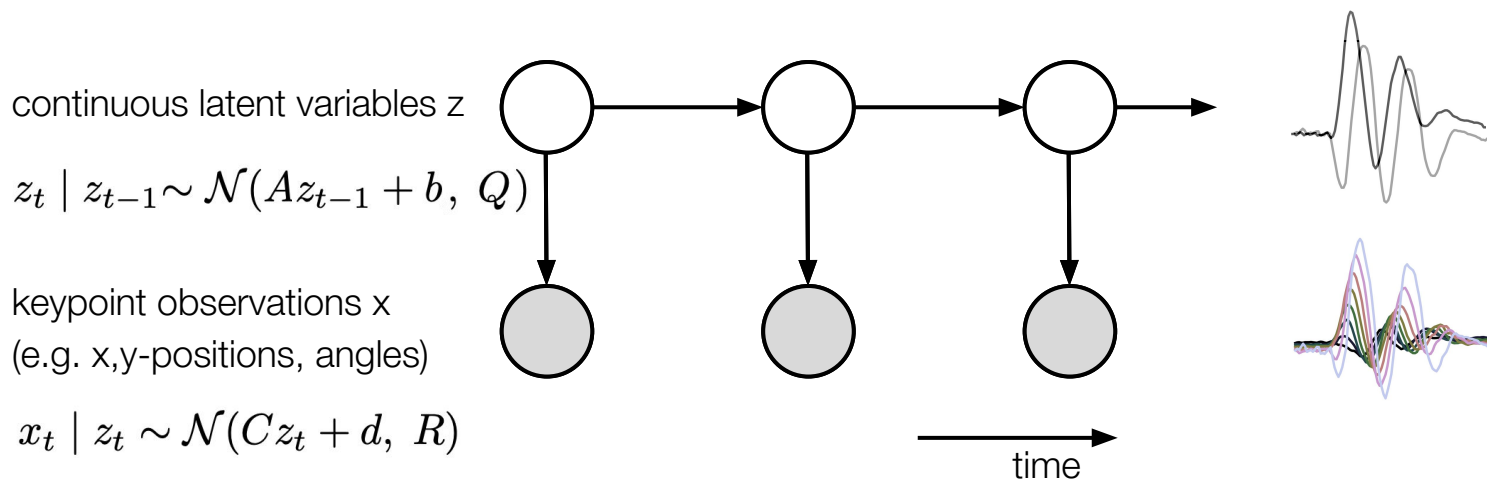
keypoint observations  $x$   
(e.g. x,y-positions, angles)



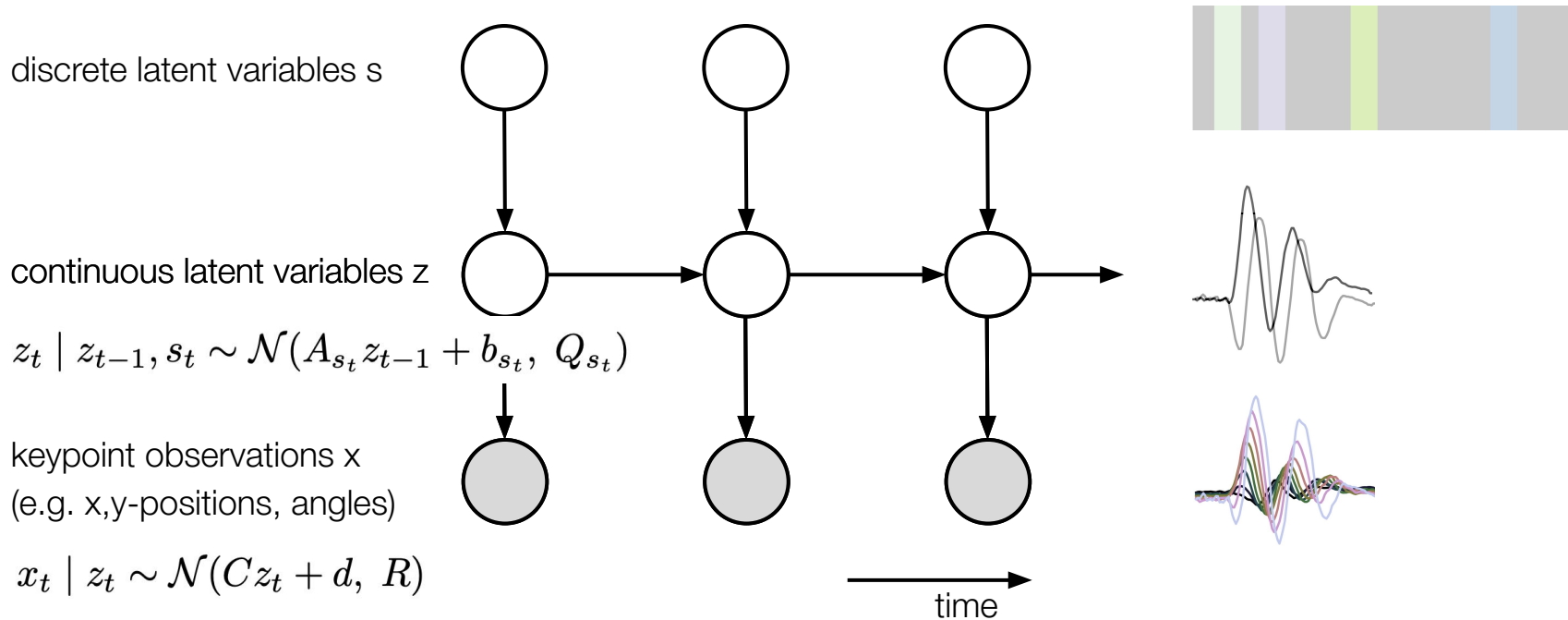
# Modeling motor time series with hierarchical structure



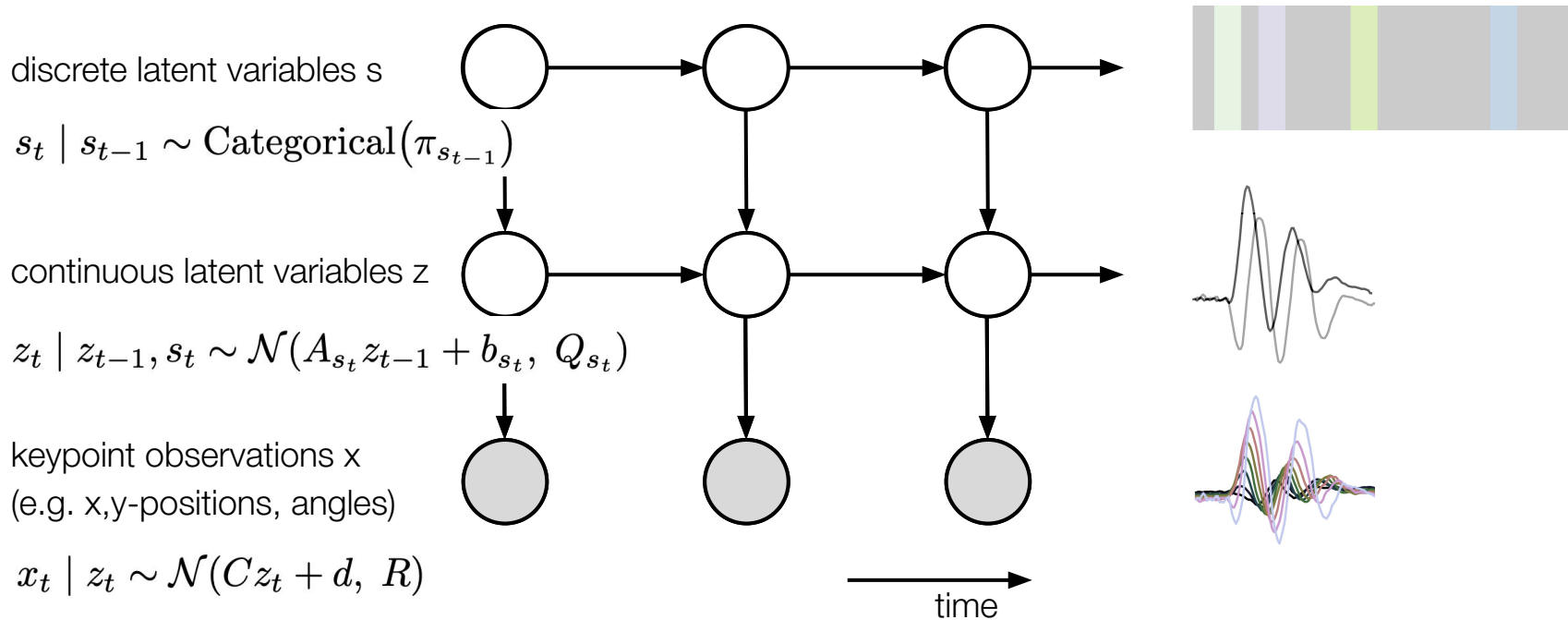
# Modeling motor time series with hierarchical structure



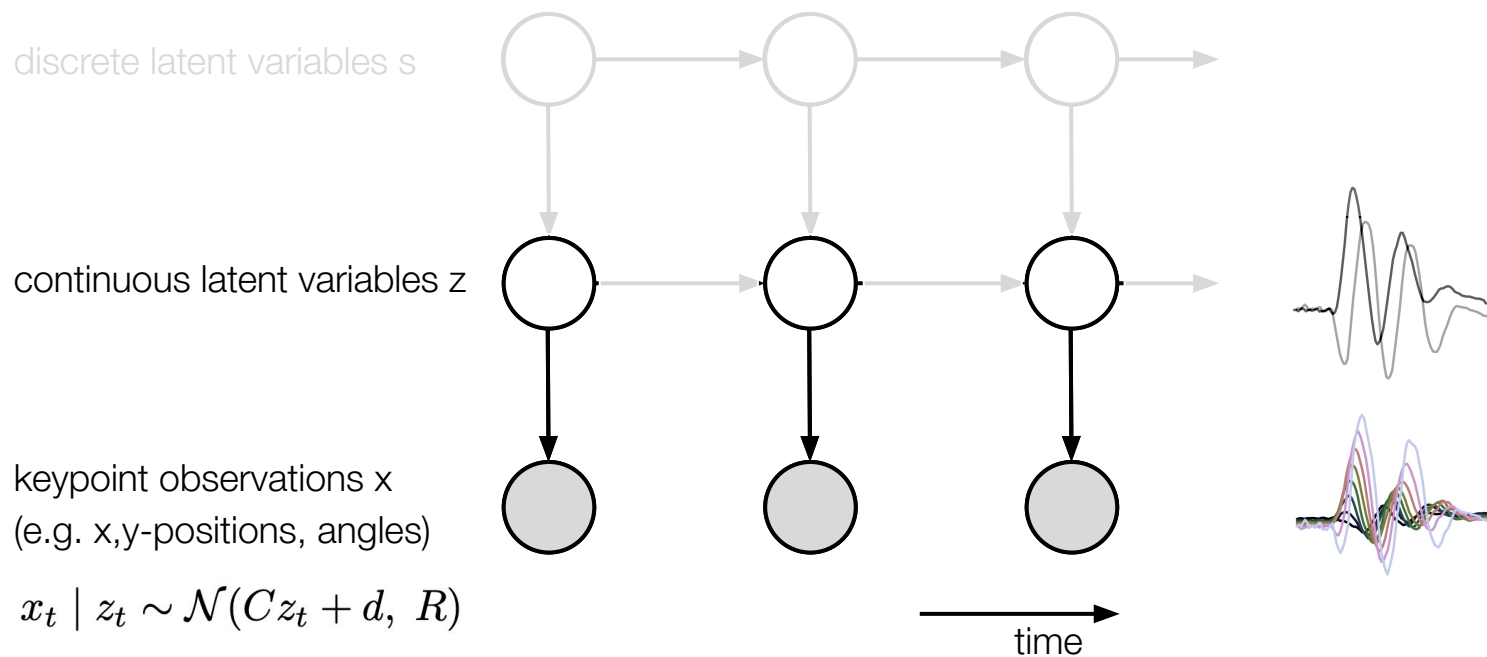
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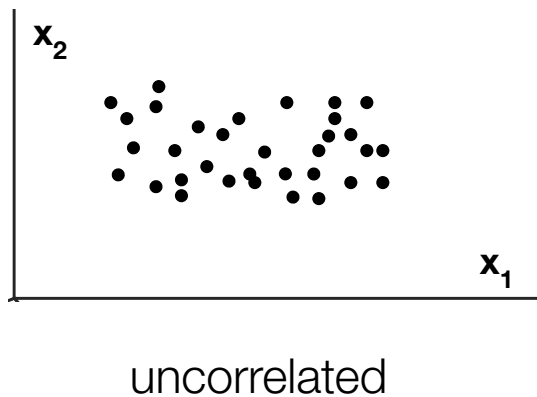
# Modeling motor time series with hierarchical structure



# Step 1: Dimensionality reduction

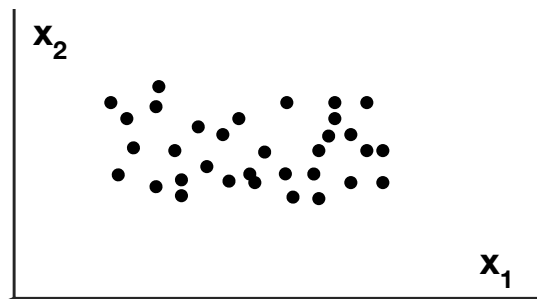


# Refresher: PCA

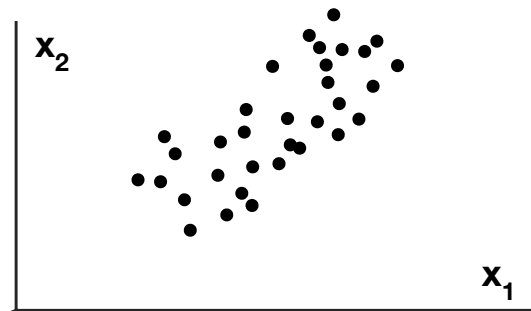




# Refresher: PCA

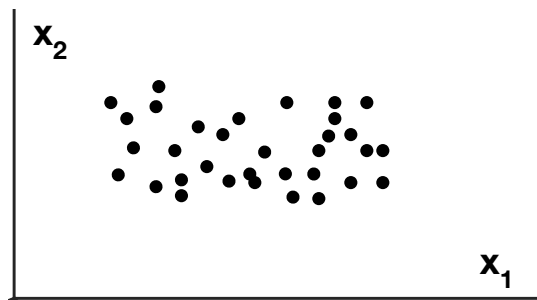


uncorrelated

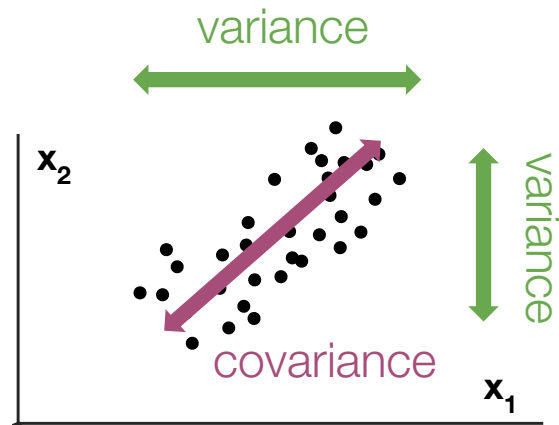


correlated

# Refresher: PCA



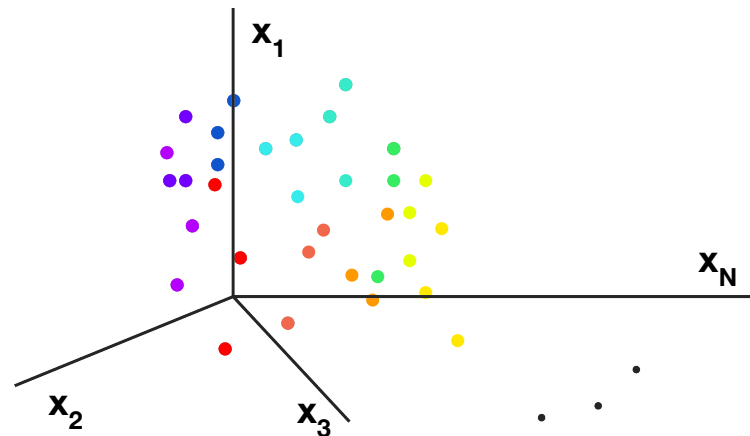
uncorrelated



correlated

# Refresher: PCA

PCA finds directions of maximal variance:



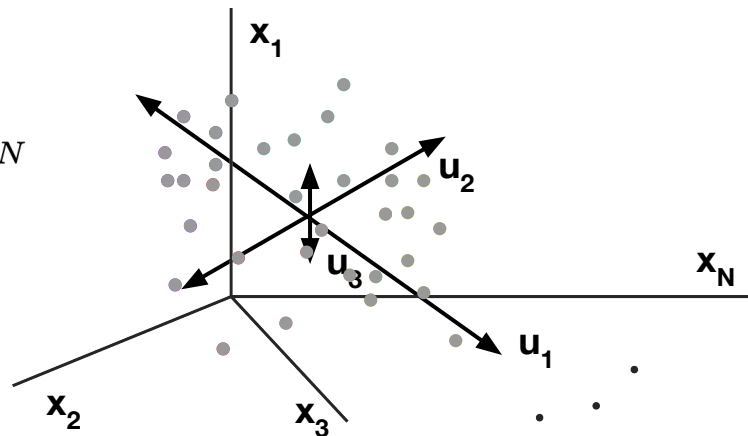
# Refresher: PCA

PCA finds directions of maximal variance:

$$\mathbf{z} = \mathbf{U}^T \mathbf{x}$$

with projections/scores  $\mathbf{z} \in R^{N \times T}$

weights/loadings/principal components  $\mathbf{U} \in R^{N \times N}$



# Refresher: PCA

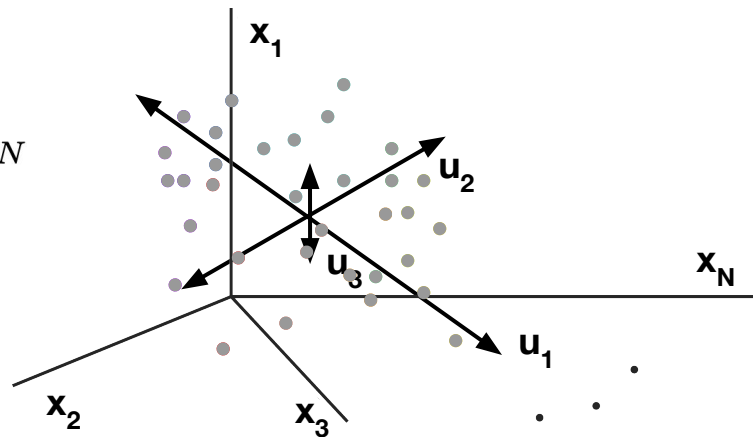
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$$\mathbf{U} = \begin{bmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \dots & \mathbf{u}_N \\ | & | & & | \end{bmatrix}$$



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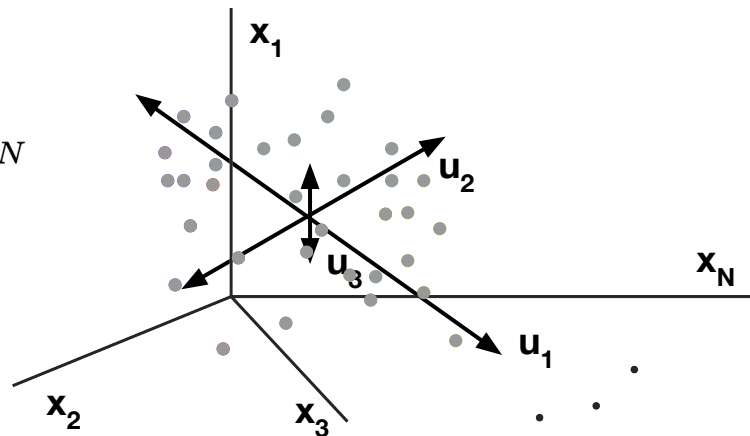
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$$\lambda_1 > \lambda_2 > \dots > \lambda_N$$

eigenvalues:  
variance along each PC



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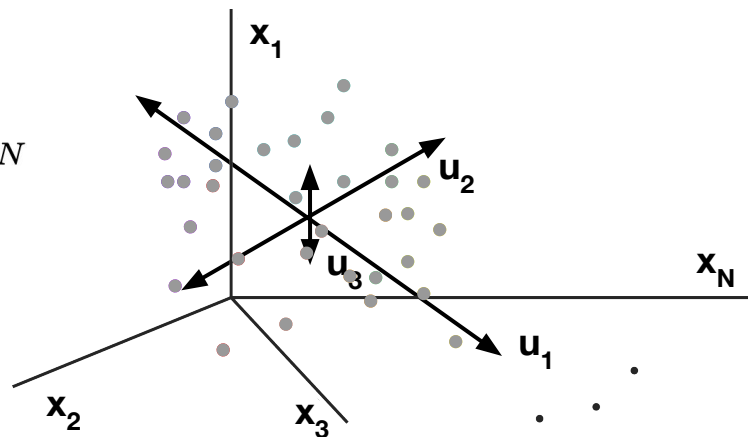
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# Refresher: PCA

PCA finds directions of maximal variance:

$$\tilde{\mathbf{z}} = \tilde{\mathbf{U}}^T \mathbf{x}$$

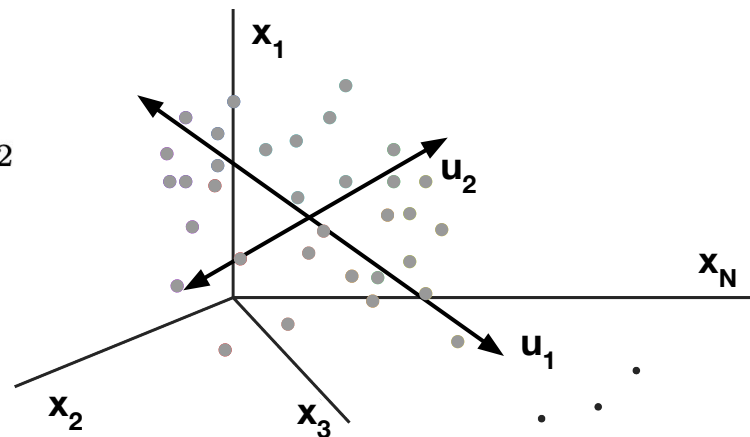
with projections/scores  $\tilde{\mathbf{z}} \in R^{2 \times T}$

weights/loadings/principal components  $\tilde{\mathbf{U}} \in R^{N \times 2}$

$$\tilde{\mathbf{U}} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix}$$

$$\lambda_1 > \lambda_2$$

eigenvalues:  
variance along each PC





# Refresher: PCA

PCA finds directions of maximal variance:

$$\tilde{\mathbf{z}} = \tilde{\mathbf{U}}^\top \mathbf{x}$$

with projections/scores  $\tilde{\mathbf{z}} \in R^{2 \times T}$

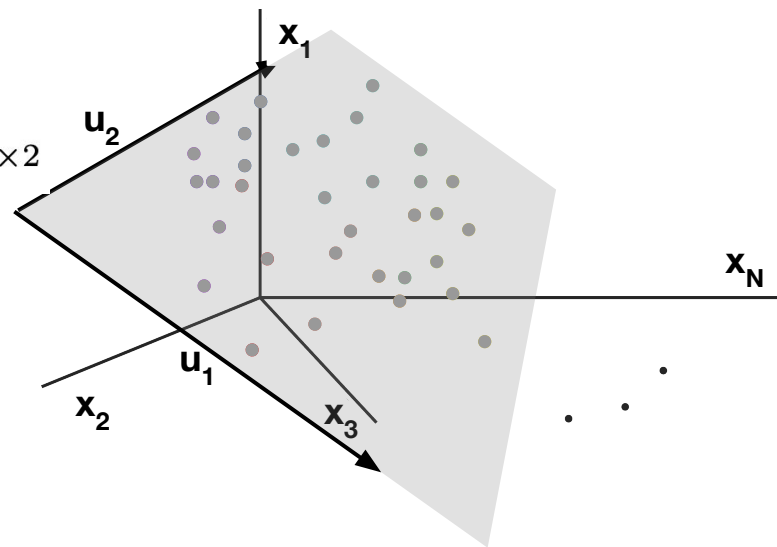
weights/loadings/principal components  $\tilde{\mathbf{U}} \in R^{N \times 2}$

$$\tilde{\mathbf{U}} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix}$$

$$\lambda_1 > \lambda_2$$

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variance along each PC

PCs  $\mathbf{U}$  form an *orthonormal basis*  
(they are orthogonal vectors with unit length)



# Refresher: PCA

Code: Part 1

PCA finds directions of maximal variance:

$$\tilde{\mathbf{z}} = \tilde{\mathbf{U}}^\top \mathbf{x}$$

with projections/scores  $\tilde{\mathbf{z}} \in R^{2 \times T}$

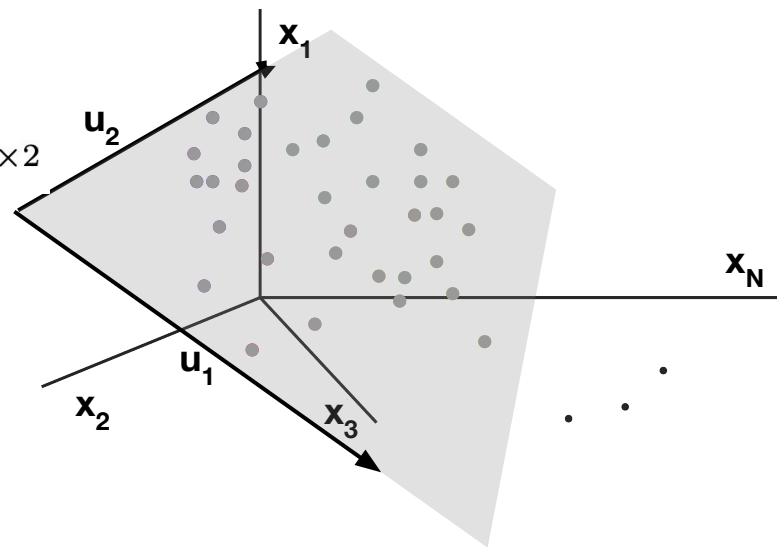
weights/loadings/principal components  $\tilde{\mathbf{U}} \in R^{N \times 2}$

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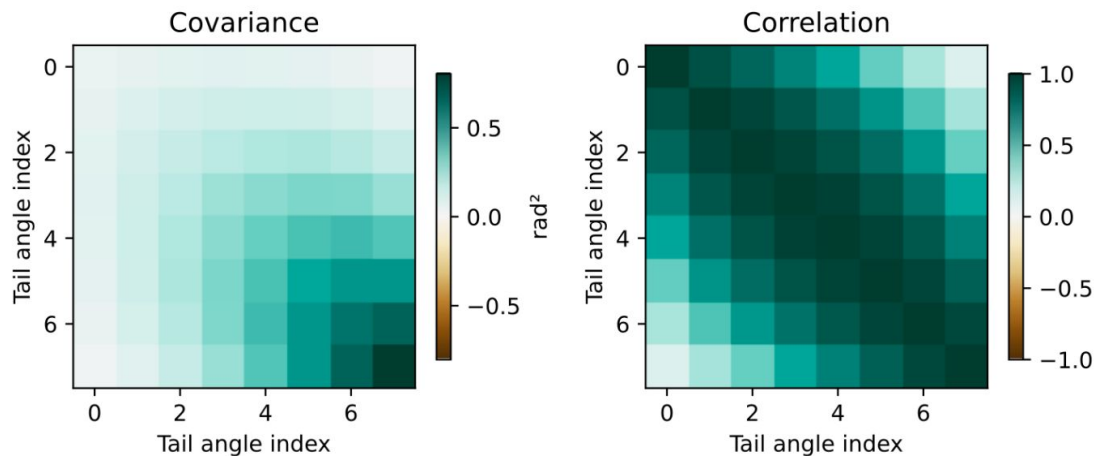
eigenvalues:  
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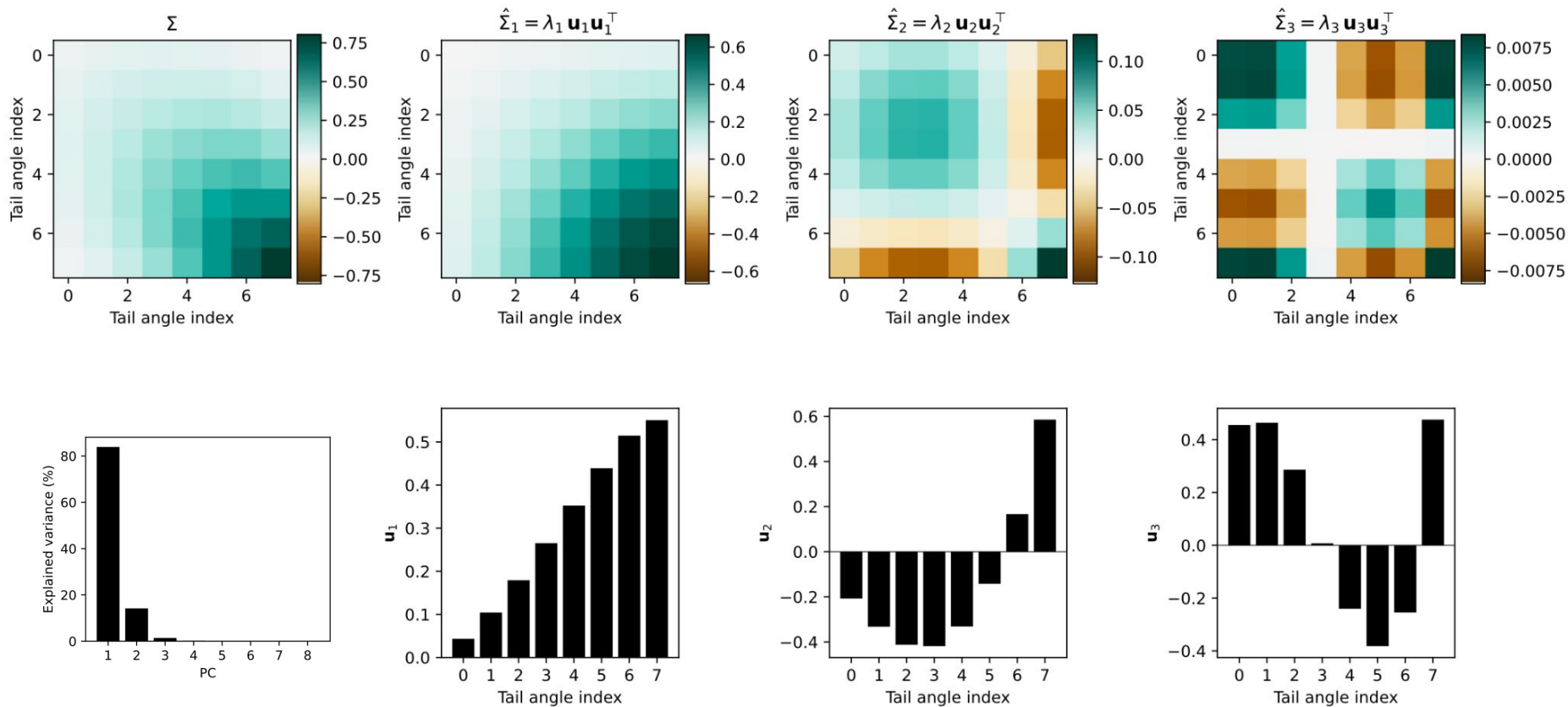




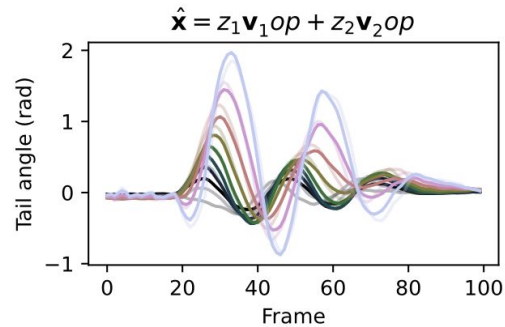
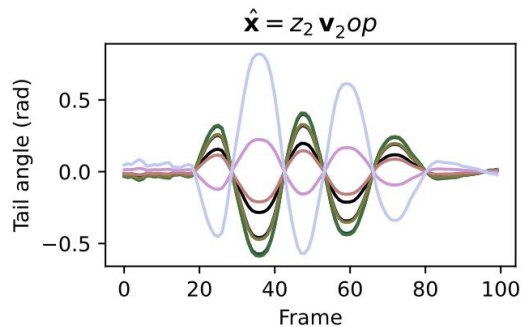
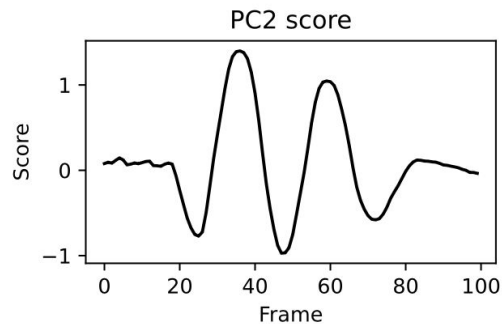
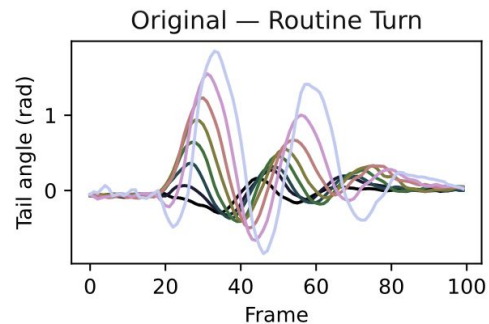
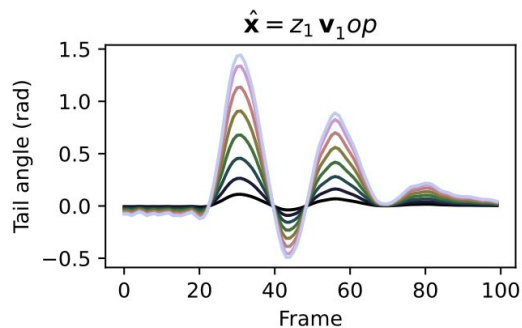
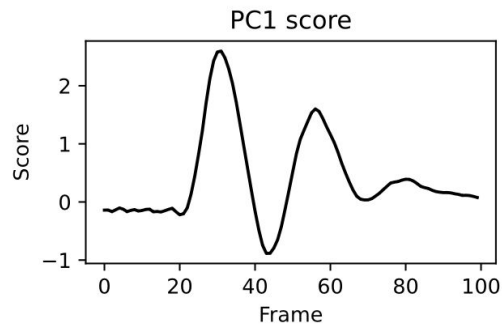
# Covariance vs. correlation



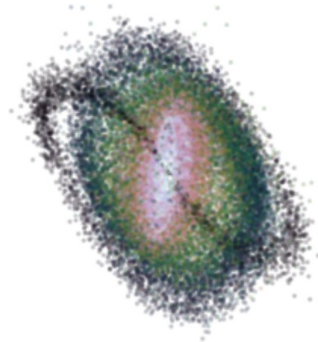
# Decomposing the covariance matrix



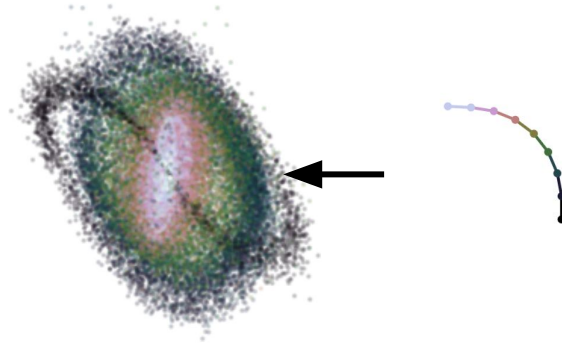
# Decomposing (& reconstructing) time series



# Trajectories in the latent space

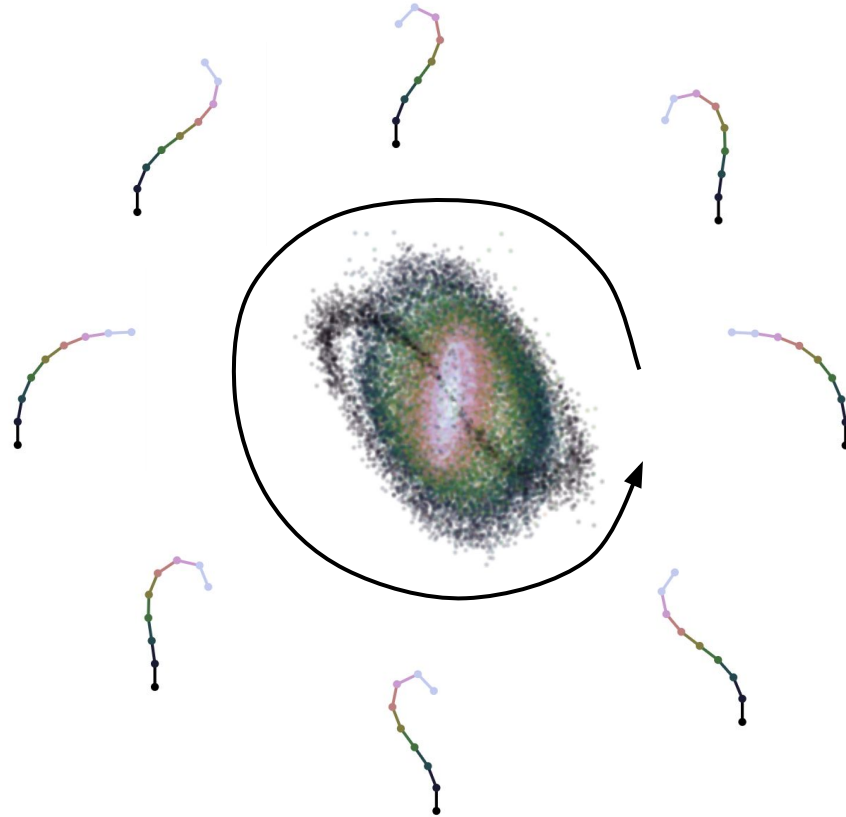


# Trajectories in the latent space



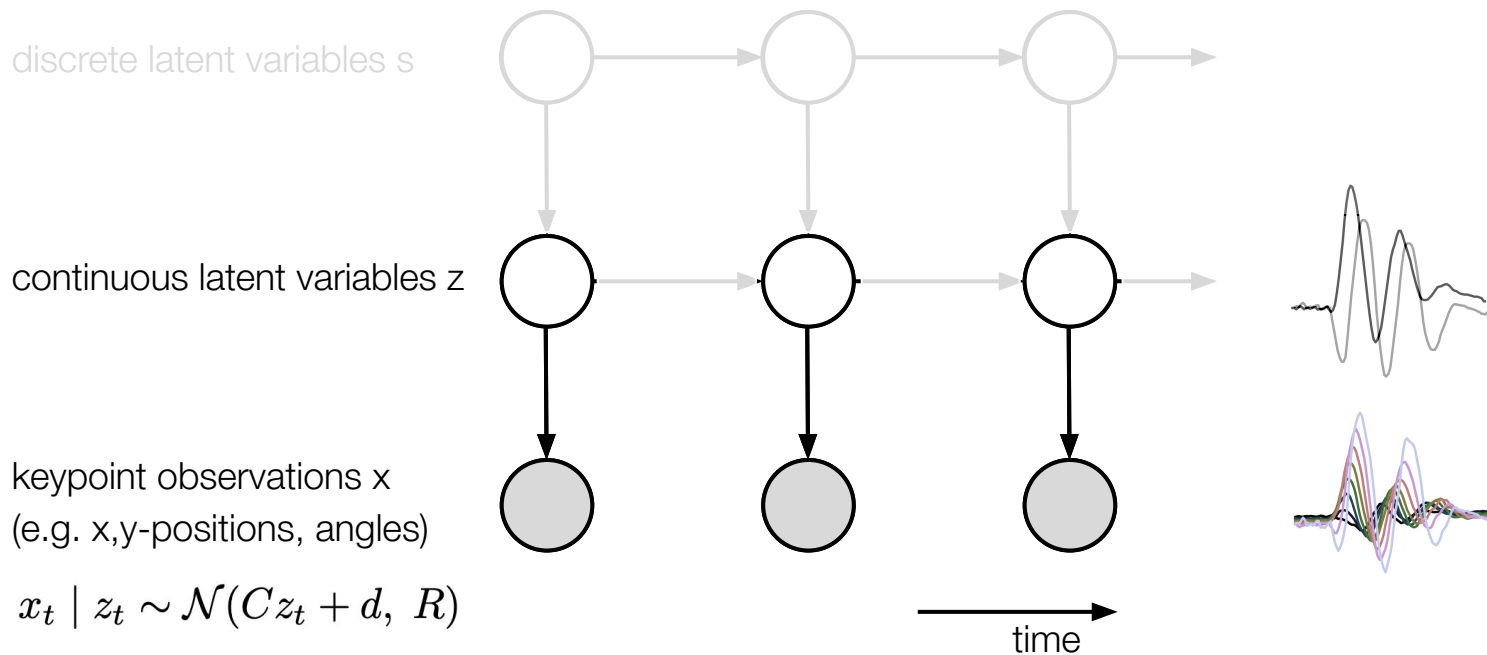


# Trajectories in the latent space

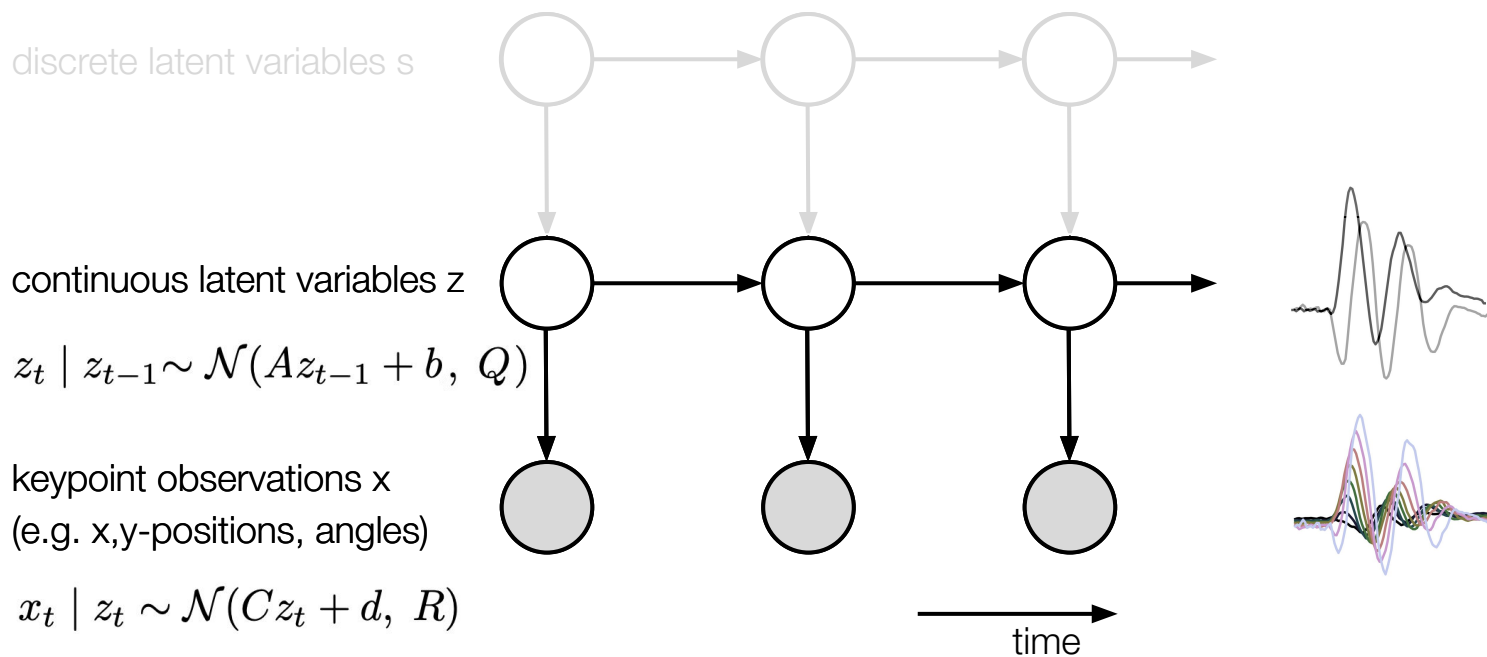




# Modeling motor time series with hierarchical structure



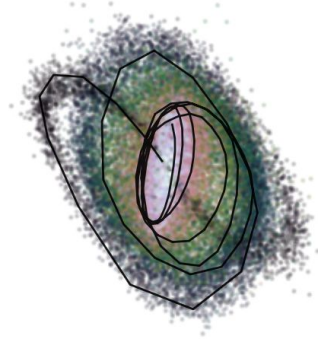
# Modeling motor time series with hierarchical structure



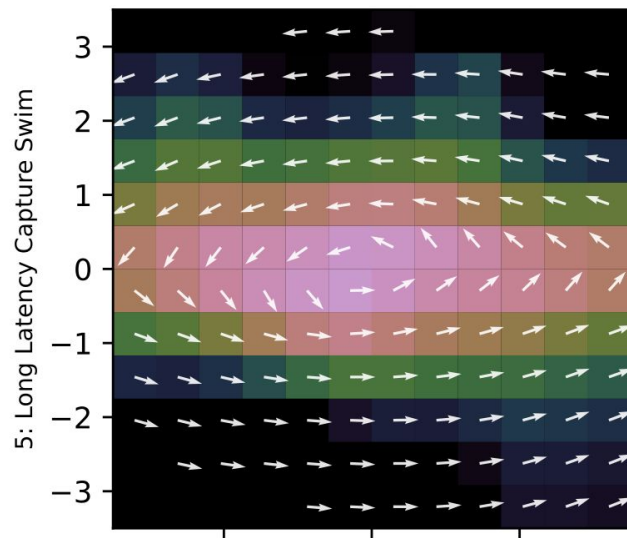
# Trajectories in the latent space



# Trajectories in the latent space



# Dynamics in latent space



# Dynamics in latent space

$A$  determines whether activity for each latent  $z$  increases or decays:

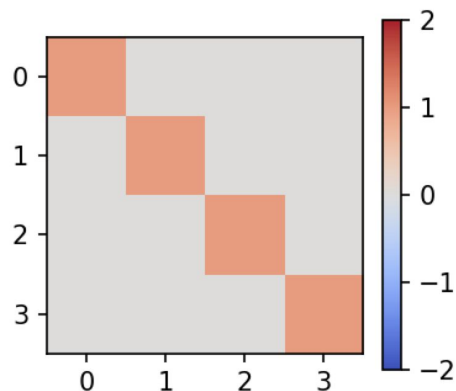
$$z_t \mid z_{t-1} \sim \mathcal{N}(Az_{t-1} + b, Q)$$



# How to understand the dynamics matrix $A$ ?

$A$  determines whether activity for each latent  $z$  increases or decays:

$$z_t \mid z_{t-1} \sim \mathcal{N}(Az_{t-1} + b, Q)$$

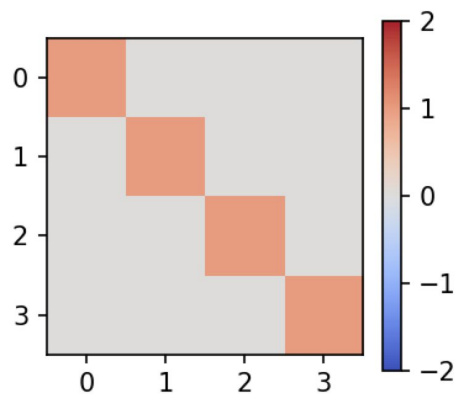


constant (no) dynamics

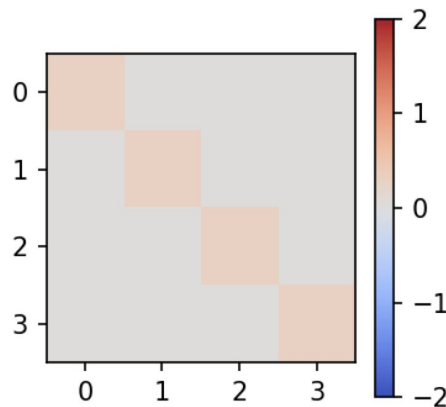
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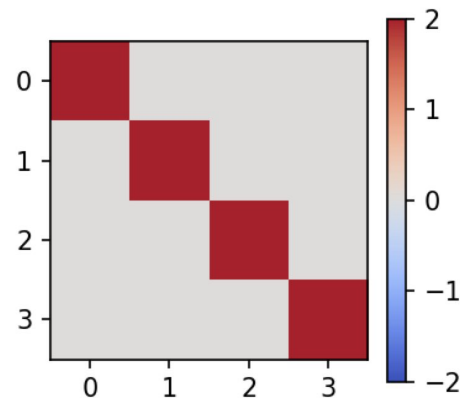
$$z_t \mid z_{t-1} \sim \mathcal{N}(Az_{t-1} + b, Q)$$



constant (no) dynamics



decaying dynamics

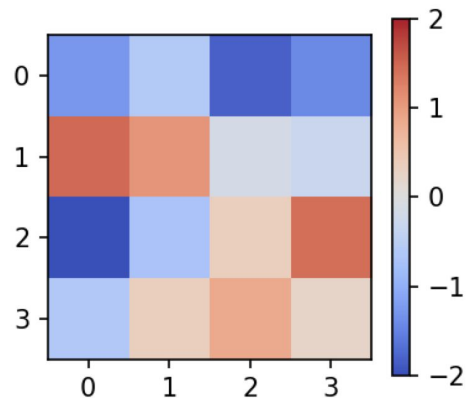


increasing dynamics

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$A$  determines whether activity for each latent  $z$  increases or decays:

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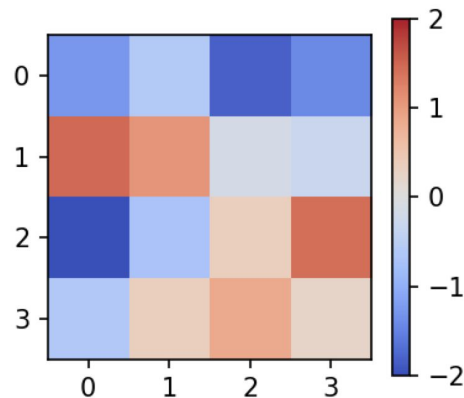


?

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$A$  determines whether activity for each latent  $z$  increases or decays:

$$z_t \mid z_{t-1} \sim \mathcal{N}(Az_{t-1} + b, Q)$$



To make sense of this matrix, we analyze its eigenvalues and the geometry between eigenvectors: Christodoulou & Vogels, 2022, OSF, <https://osf.io/preprints/osf/evqhy>

# How to understand the dynamics matrix $A$ ?

We transform  $\mathbf{A}$  linearly to a coordinate system spanned by its eigenvectors  $\mathbf{V}$ :

$$\mathbf{\Lambda} = \mathbf{V}^{-1}\mathbf{A}\mathbf{V}$$

The transformed dynamics matrix  $\mathbf{\Lambda}$  is diagonal, with eigenvalues  $\lambda$  on the diagonal.

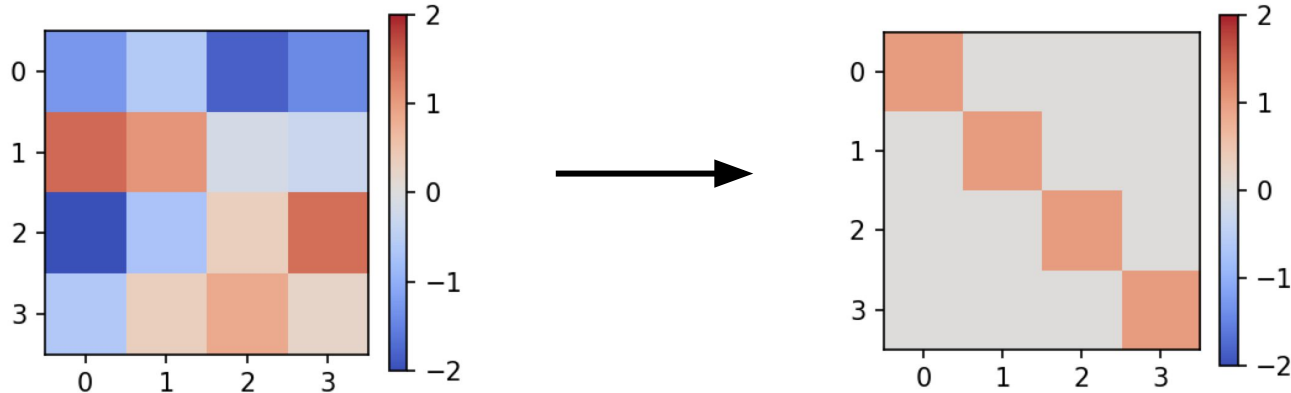
# How to understand the dynamics matrix $A$ ?

Code: Part 2

We transform  $A$  linearly to a coordinate system spanned by its eigenvectors  $V$ :

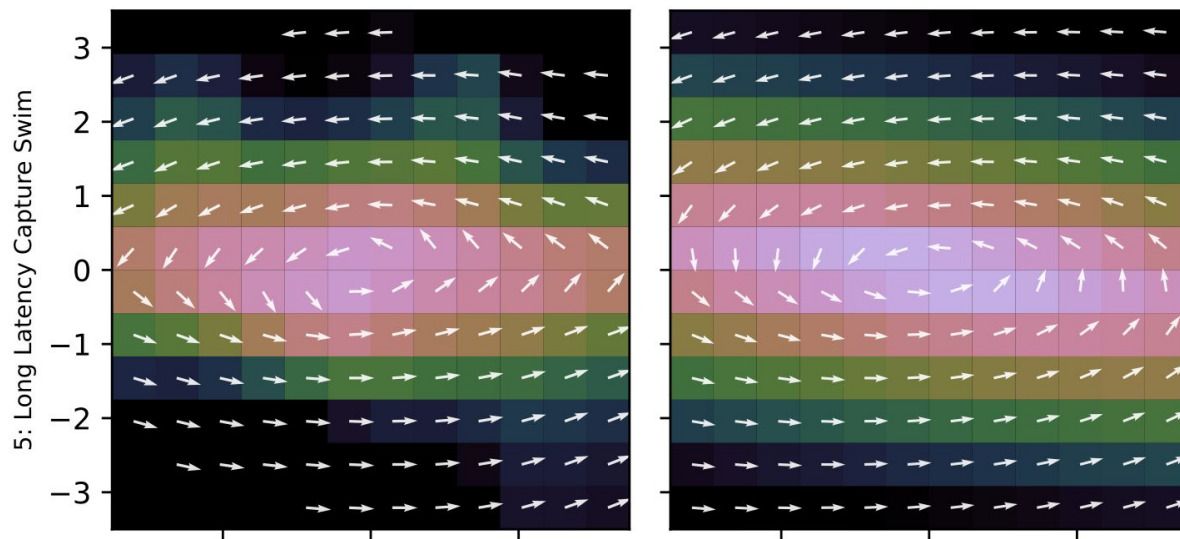
$$\Lambda = V^{-1}AV$$

The transformed dynamics matrix  $\Lambda$  is diagonal, with eigenvalues  $\lambda$  on the diagonal.



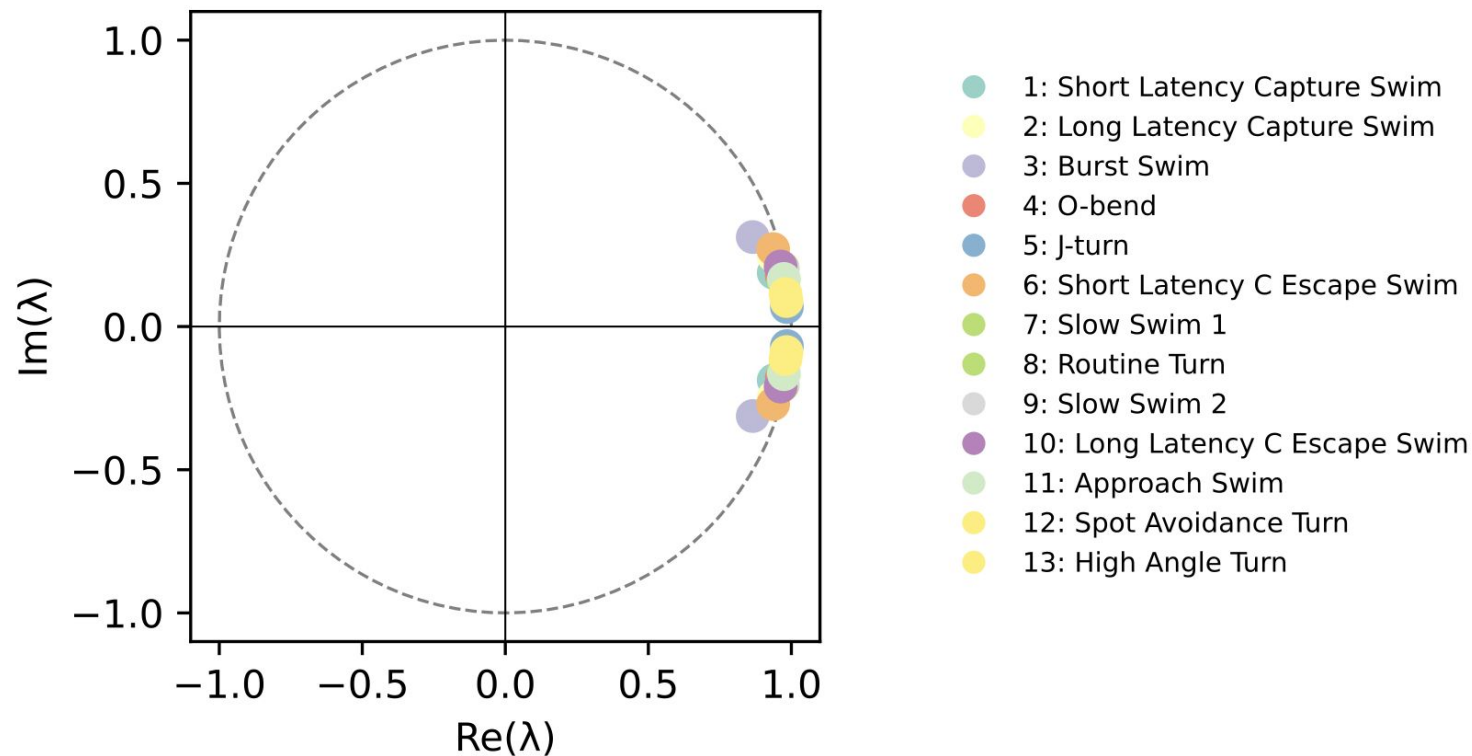


# Empirical vs fitted dynamics

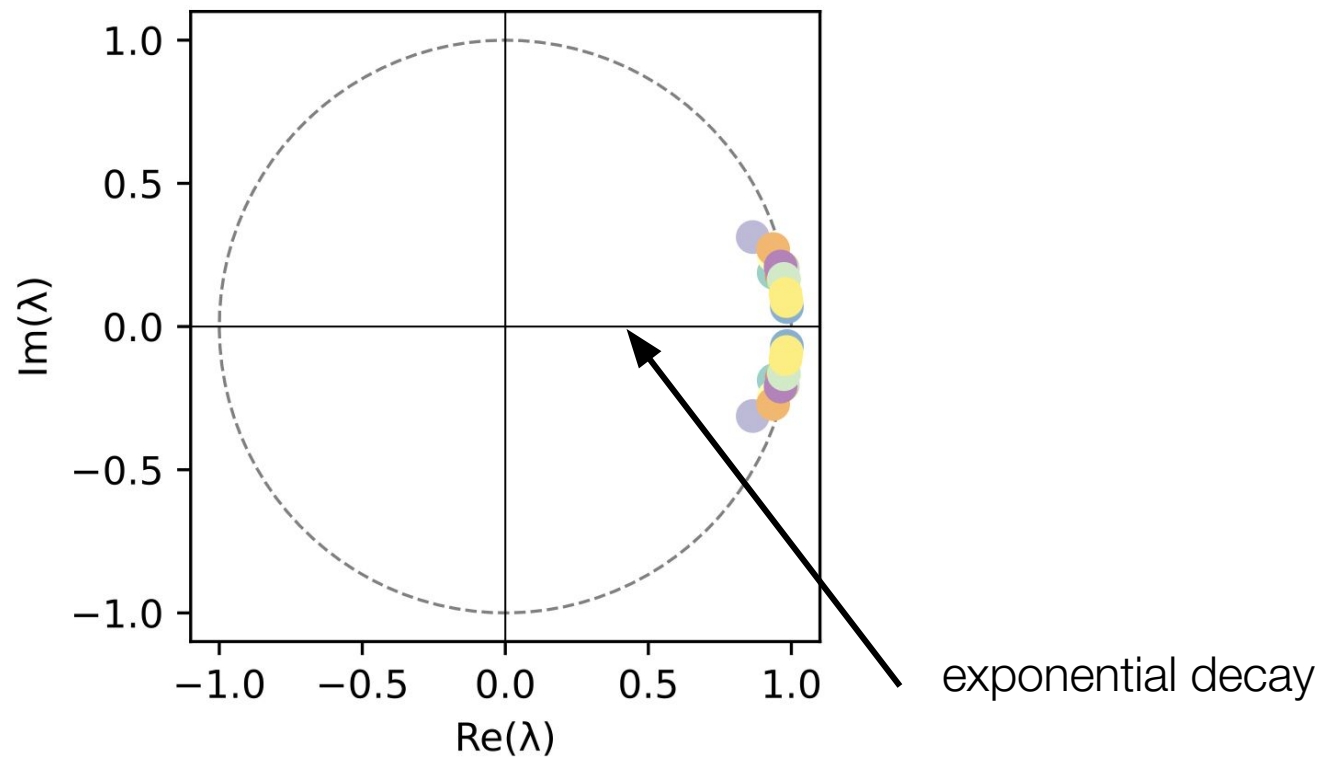




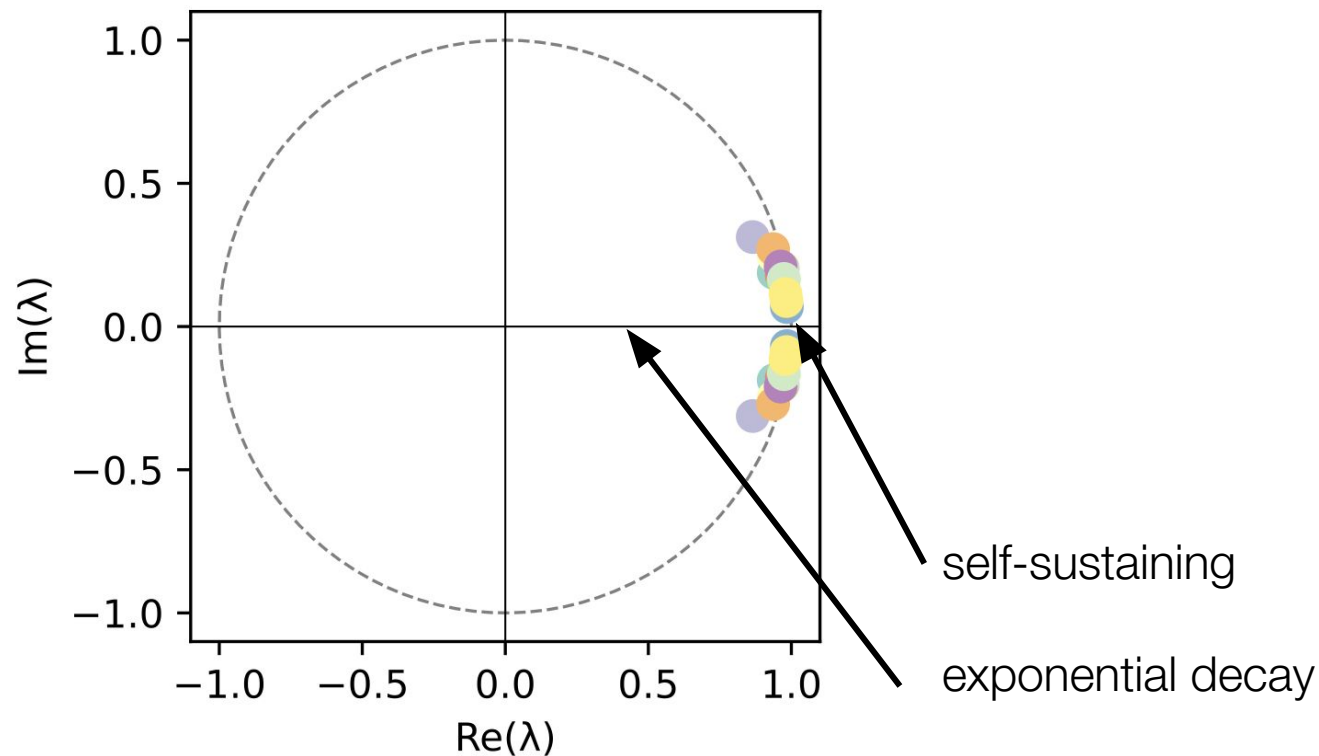
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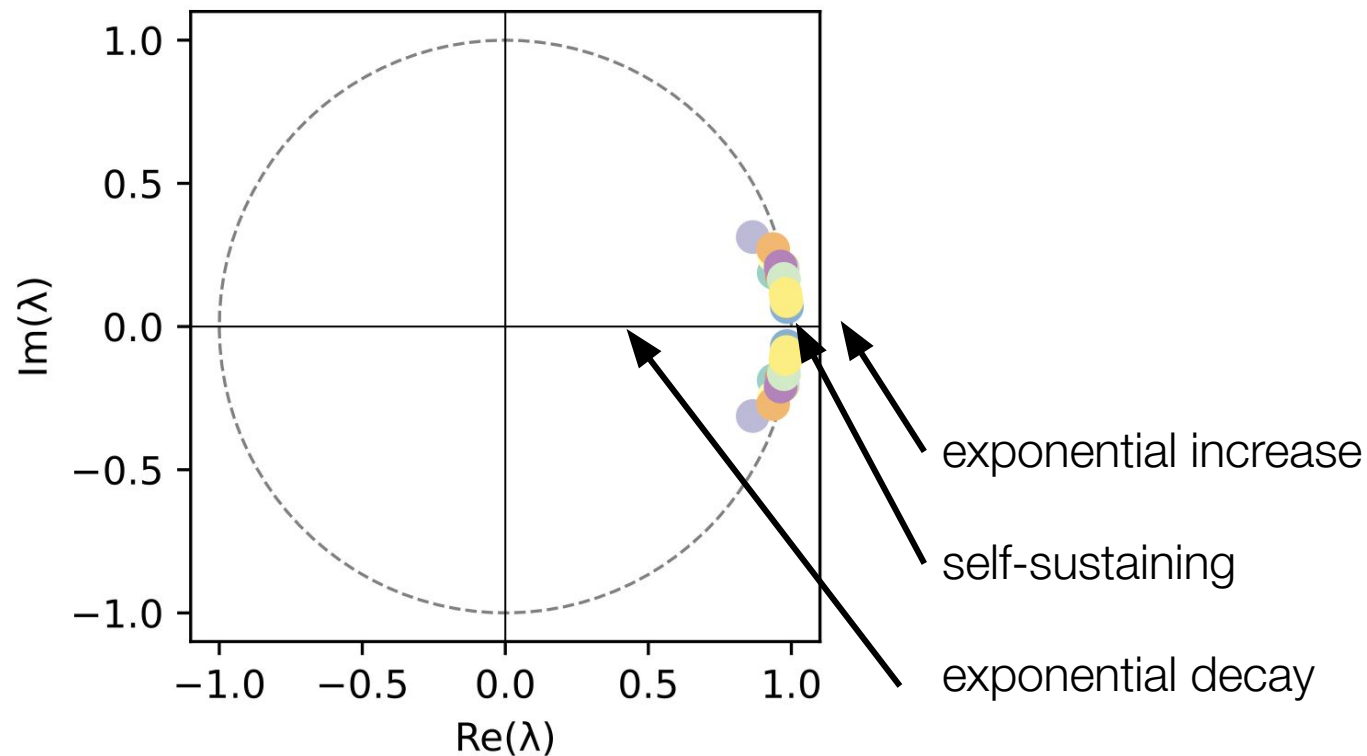
# How to understand the dynamics matrix $A$ ?



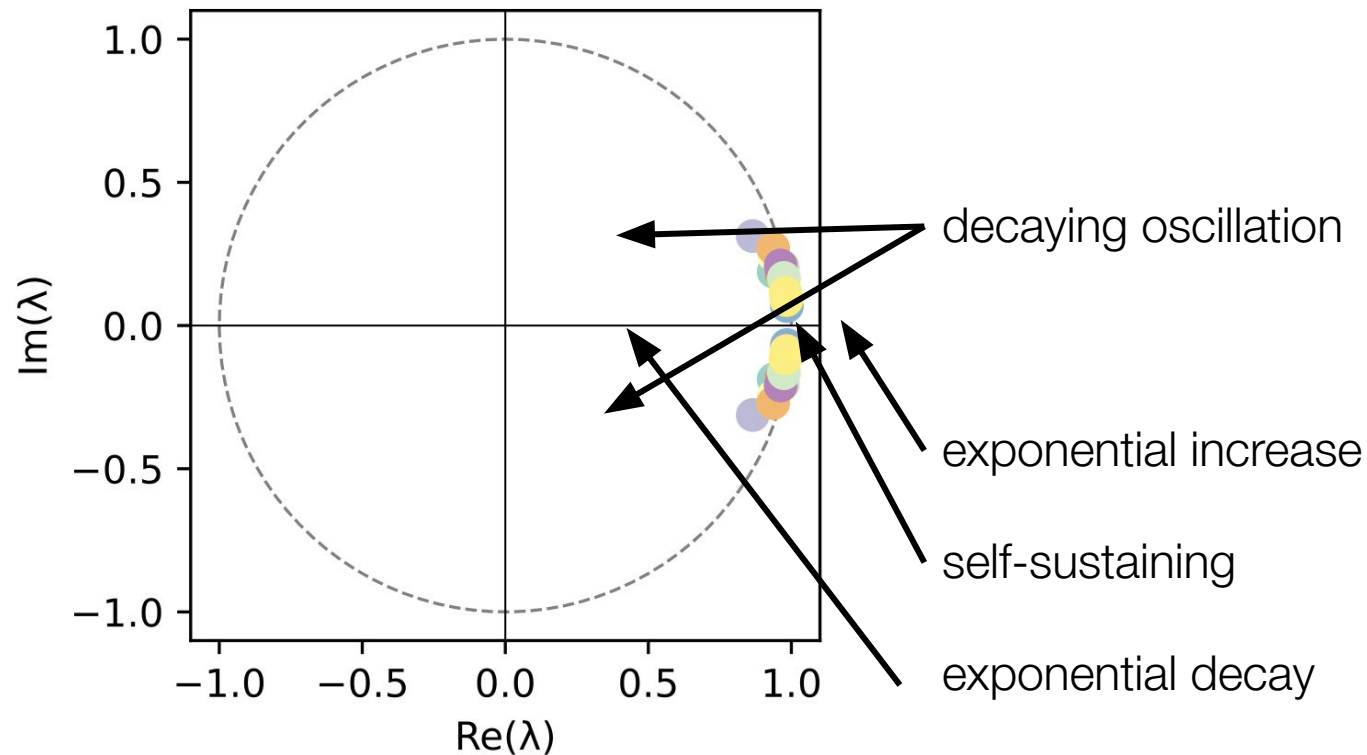
# How to understand the dynamics matrix $A$ ?



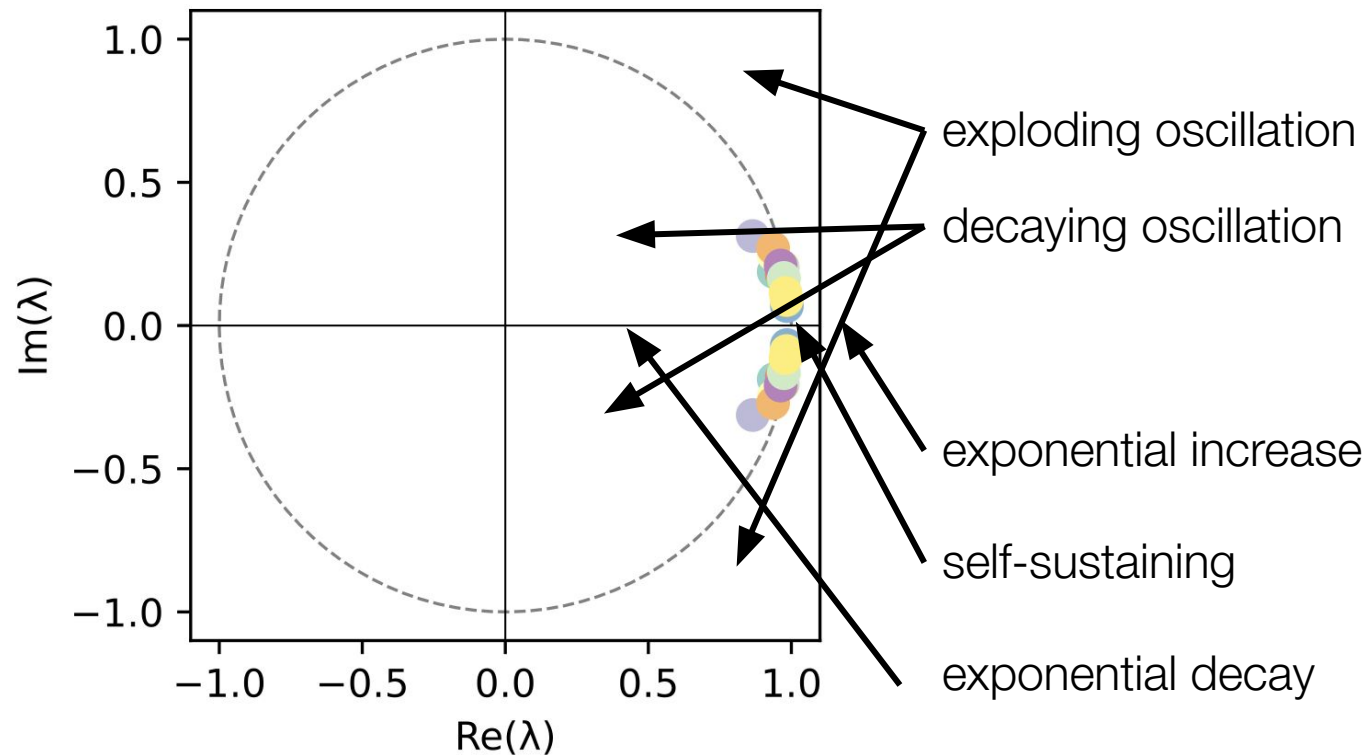
# How to understand the dynamics matrix $A$ ?



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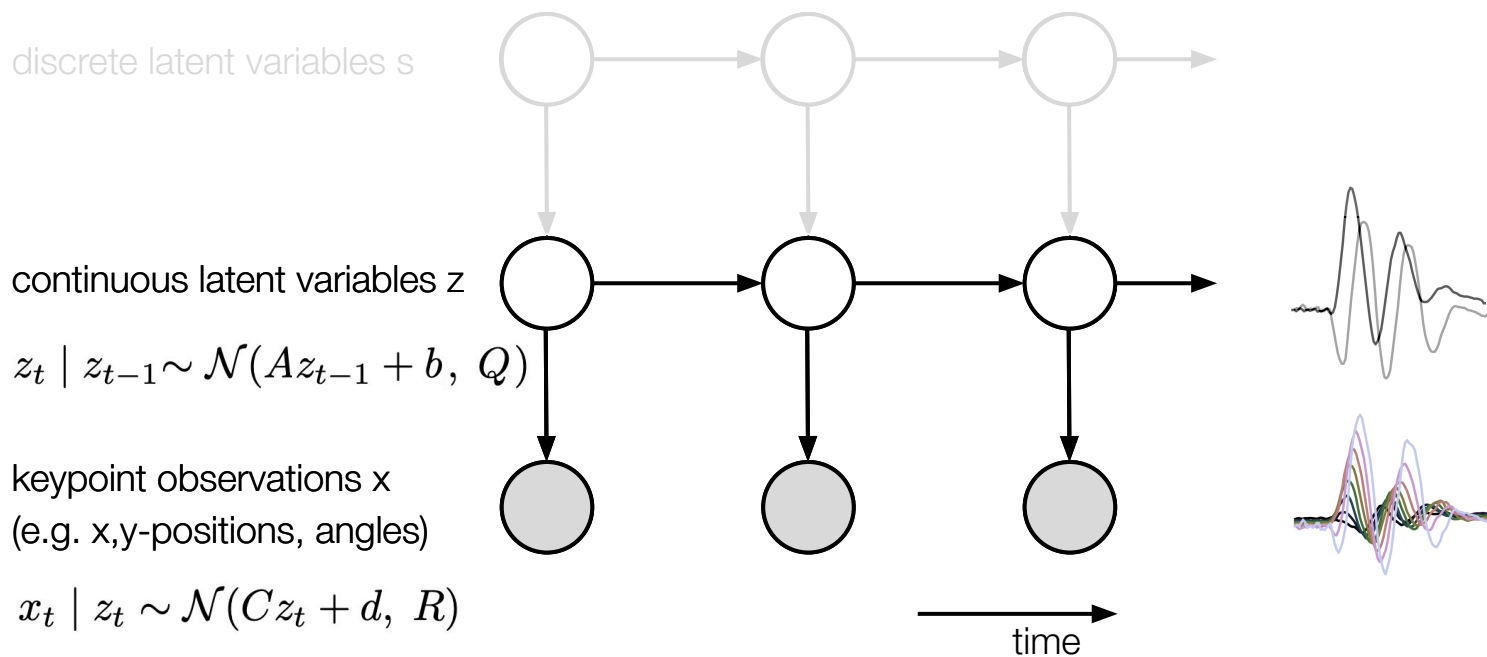


# How to understand the dynamics matrix $A$ ?



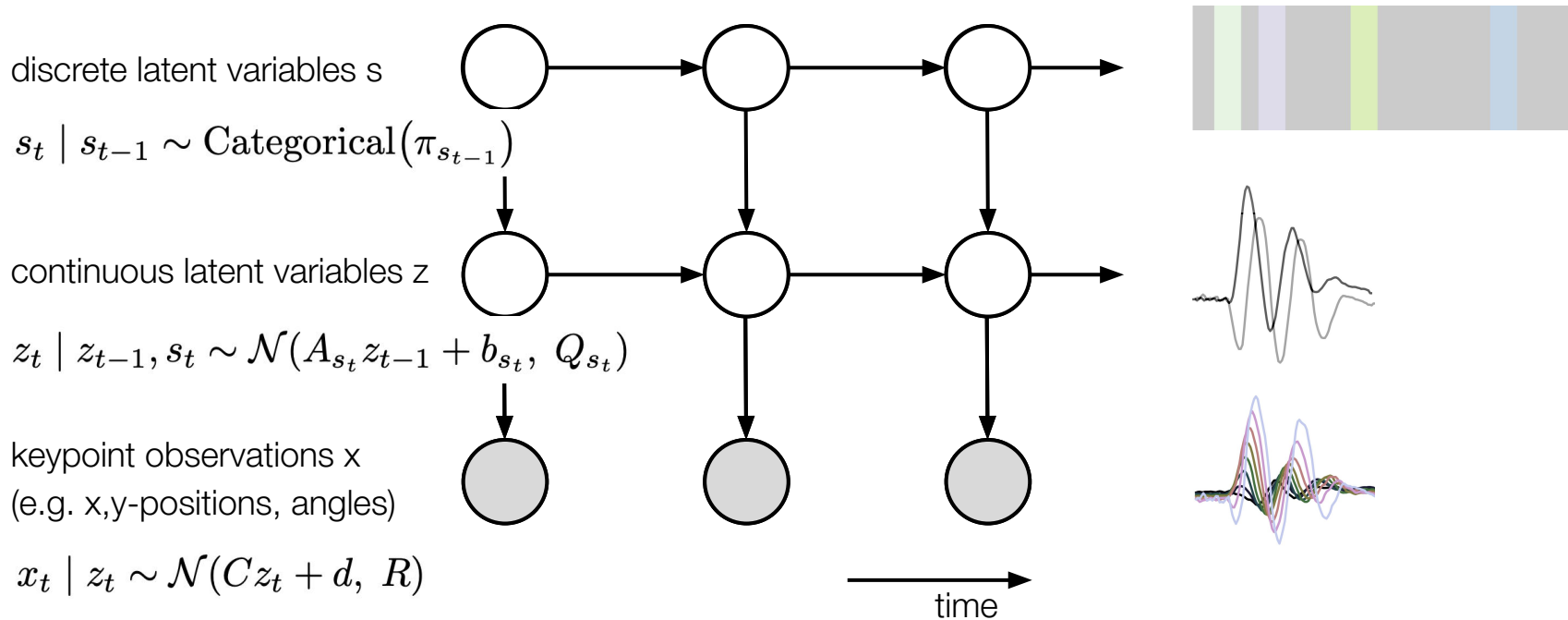


# Modeling motor time series with hierarchical structure

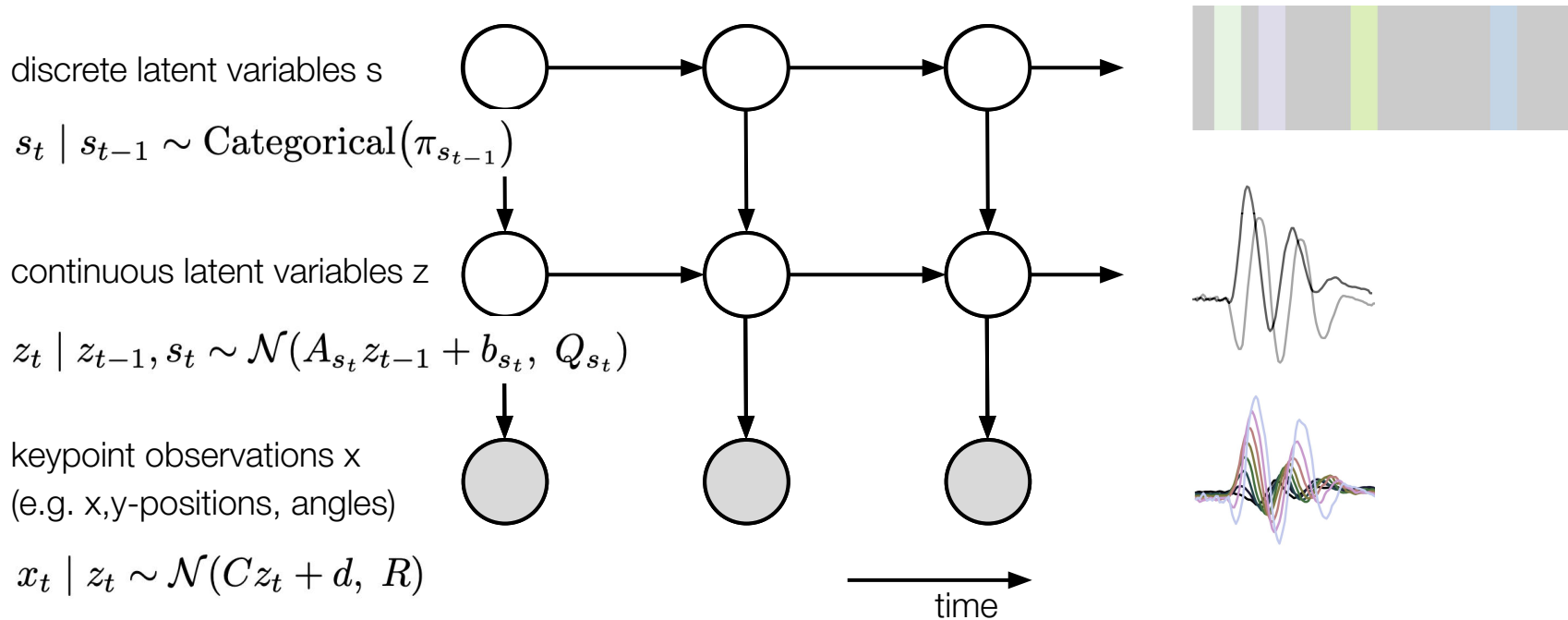




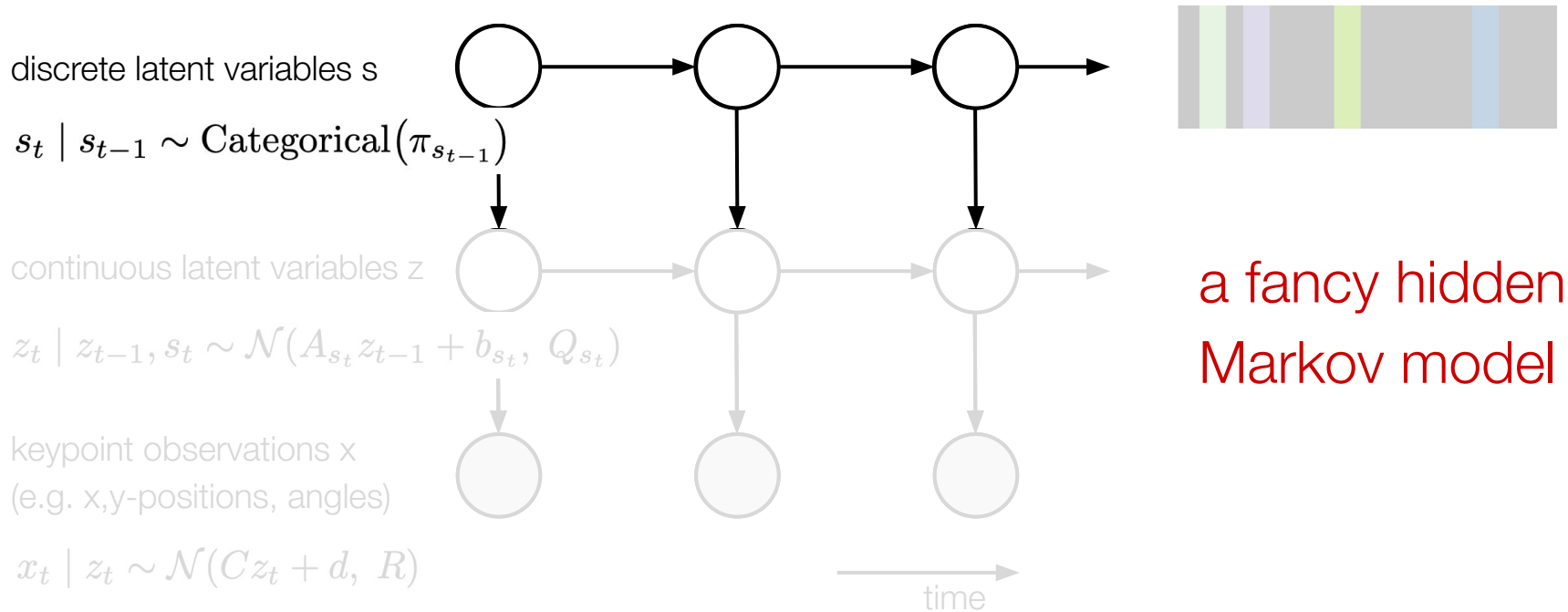
# Modeling motor time series with hierarchical structure



# Modeling motor time series with hierarchical structure



# Modeling motor time series with hierarchical structure



# HMMs in 3 minutes

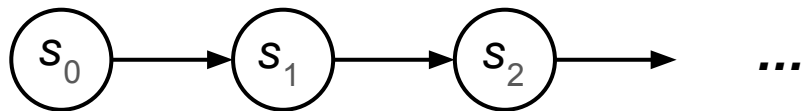
We need an **indicator variable** pointing to the syllable identity at each time point (“state”)



These states are “latent”: They are not observed, but inferred (= learned) from the data

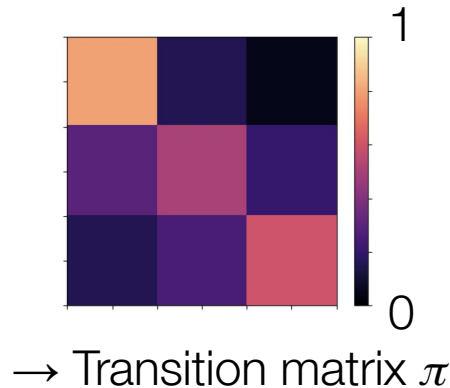
# HMMs in 3 minutes

We need an **indicator variable** pointing to the syllable identity at each time point (“state”)



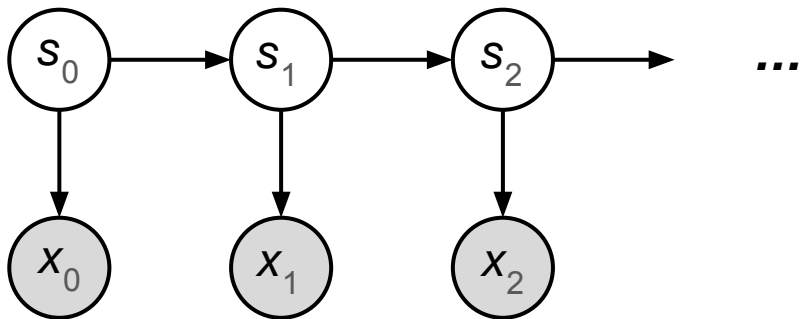
These states are “latent”: They are not observed, but inferred (= learned) from the data

We need a **model of state transitions** to capture temporal structure in discrete syllables. State transitions are probabilistic and depend only on the previous state (“Markov chain”)



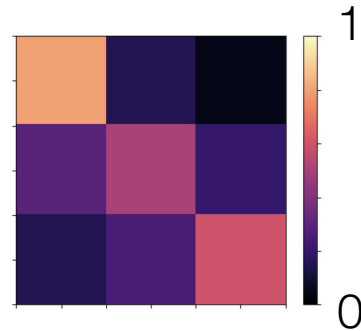
# HMMs in 3 minutes

We need an **indicator variable** pointing to the syllable identity at each time point (“state”)



We need a **model of state transitions**: The Markov chain

Each syllable has different movement dynamics: Observations depend on the state through different **emission models**



# HMMs in 120 minutes

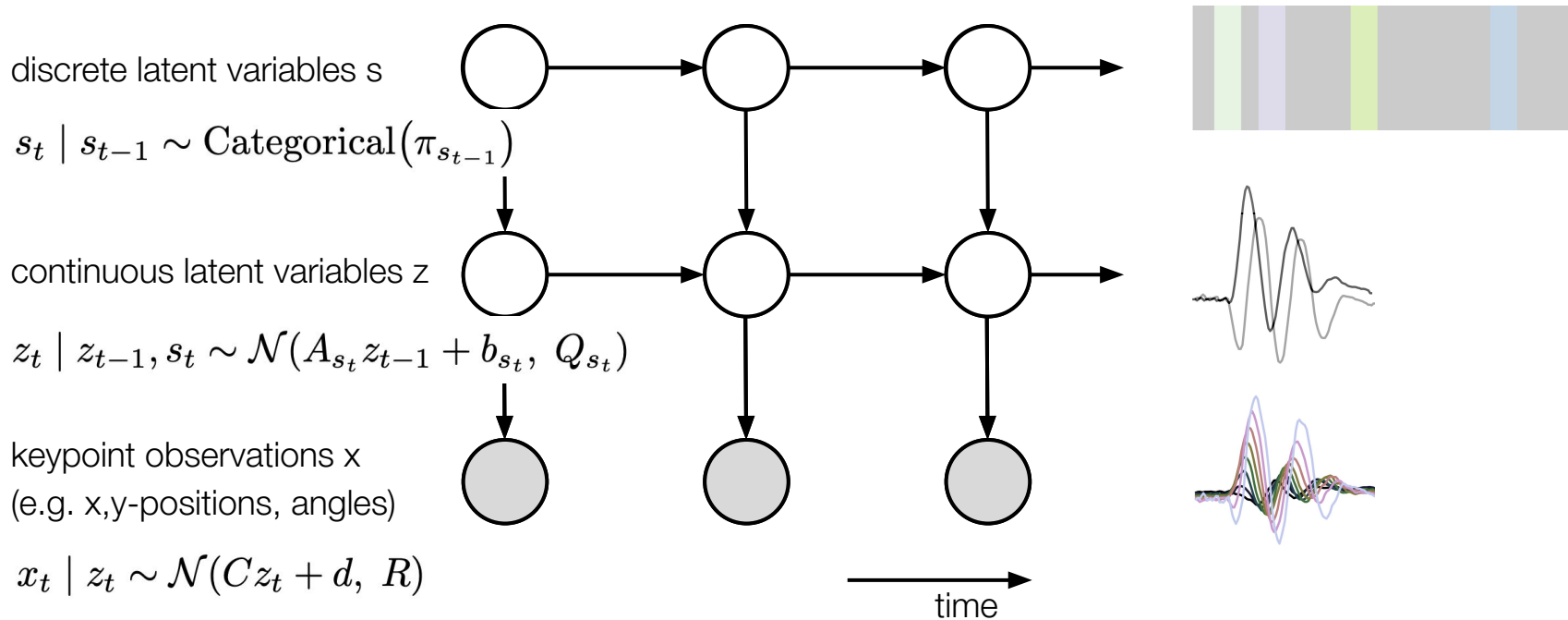
Learn the theory behind (Mixture Models and) Hidden Markov Models:

[https://github.com/bambschool/BAMB2024/blob/main/slides\\_lectures/Latent\\_variable\\_models.pdf](https://github.com/bambschool/BAMB2024/blob/main/slides_lectures/Latent_variable_models.pdf)

Implement a (Mixture Model and) a Hidden Markov Model:

<https://github.com/bambschool/BAMB2024/tree/main/Day%203A%20Latent%20variable%20models>

# Modeling motor time series with hierarchical structure





# Model hierarchy

**Code: Part 3**

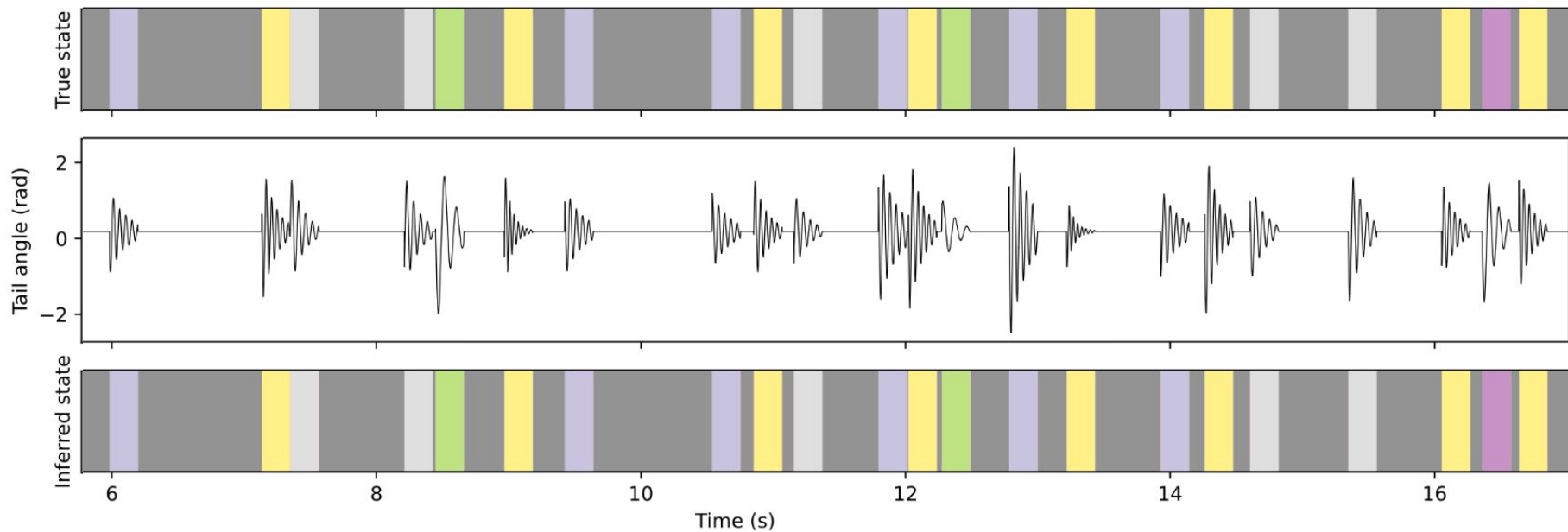
**HMM**: switches between discrete behavioral states and gives state-specific observations.

**AR-HMM**: An HMM where observations have autoregressive dynamics.

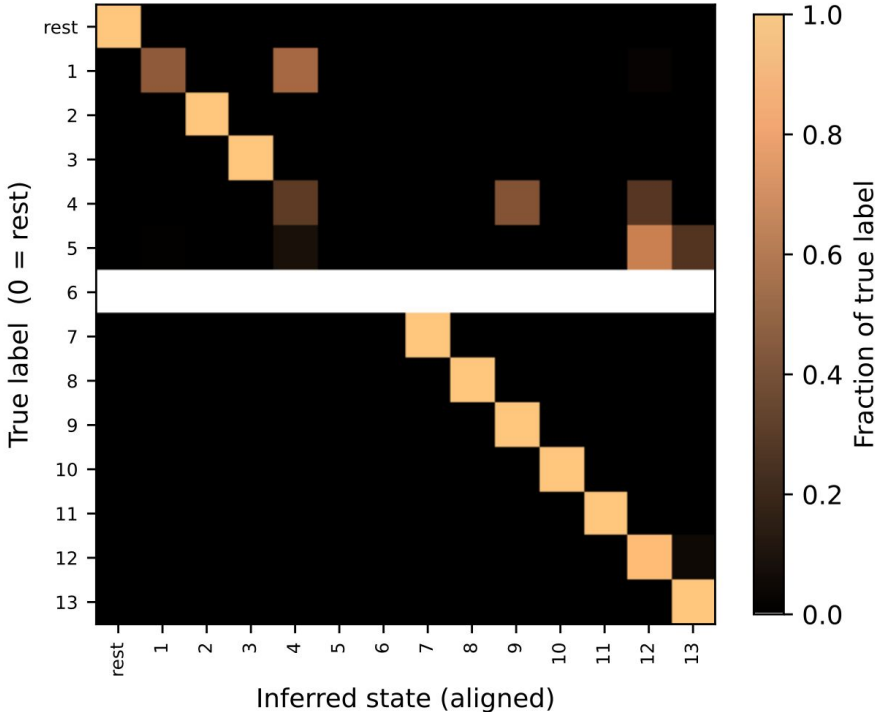
**sLDS**: An HMM with latent autoregressive dynamics. Observations are a high-dimensional linear projection of the latents.

→ All implemented in the **ssm toolbox**

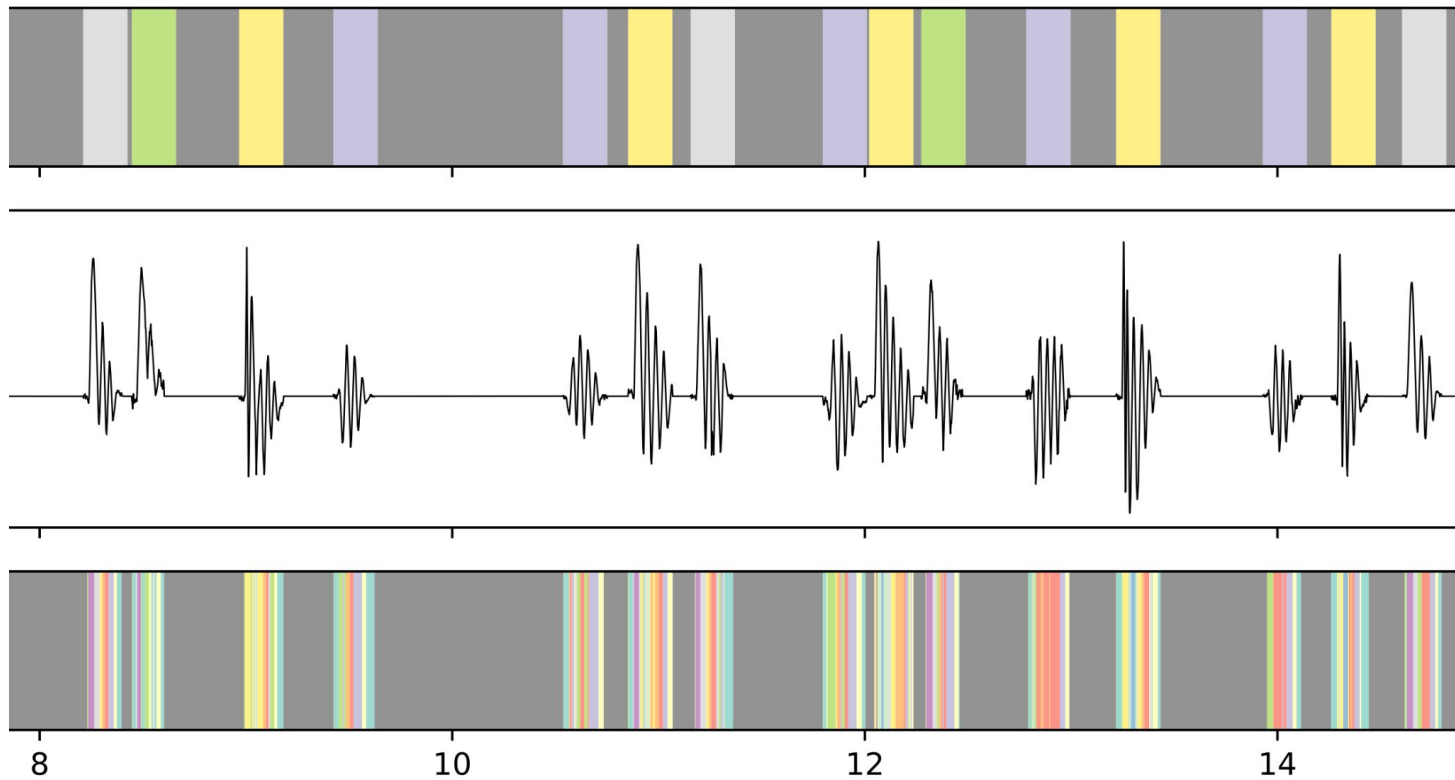
# State inference



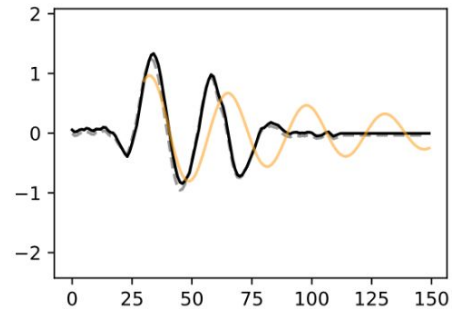
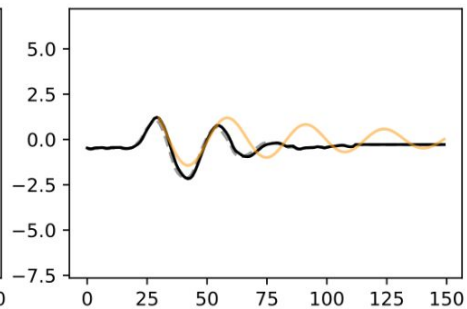
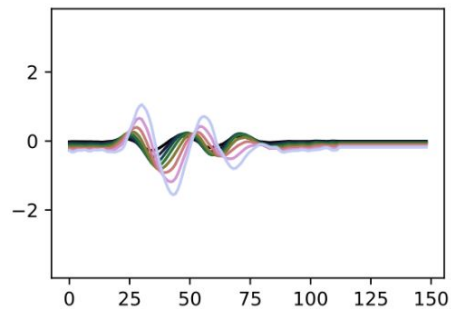
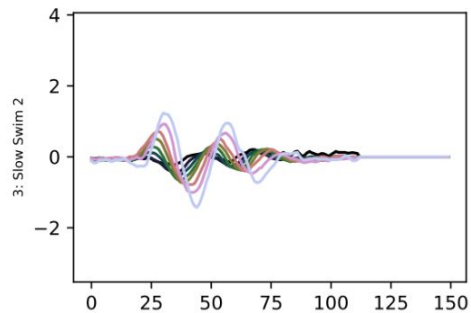
# State inference



# State inference



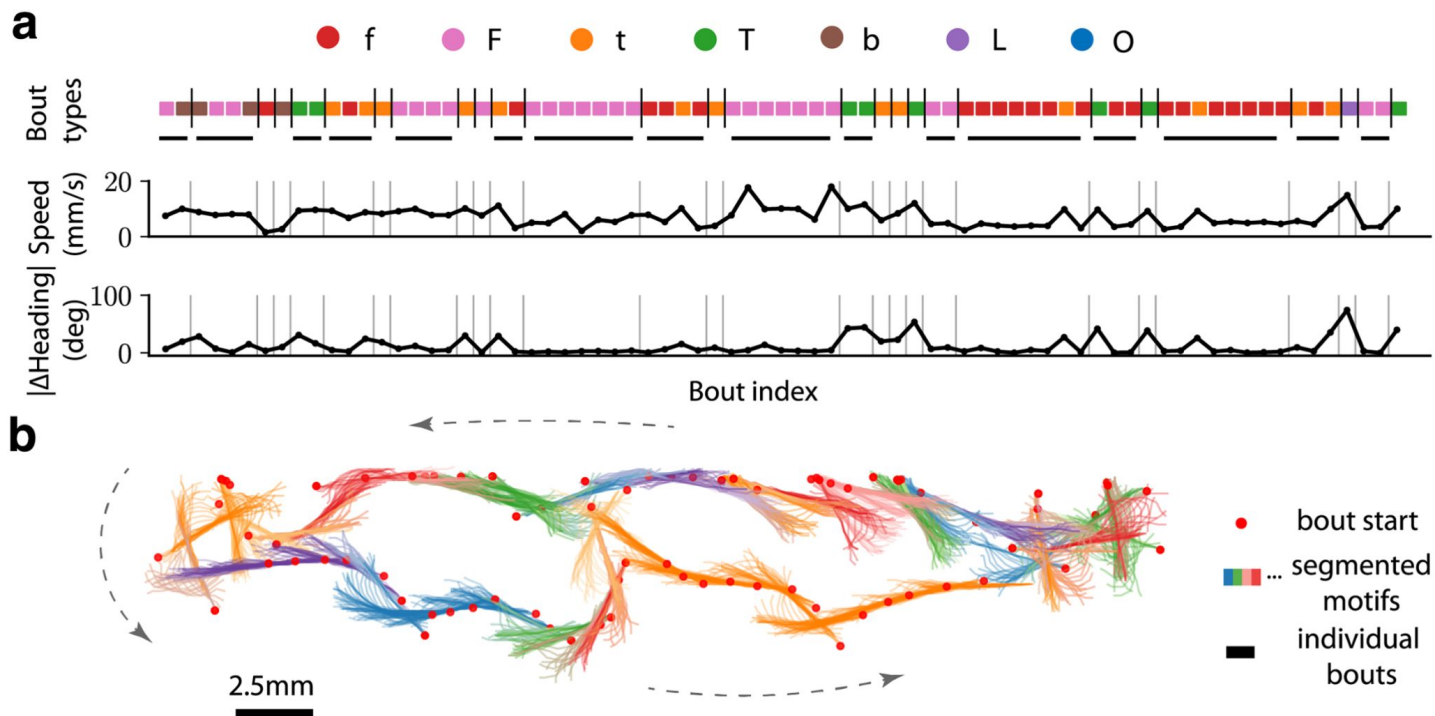
# Dynamics rollout



# Limitations of LDS and sLDS

- LDS can only capture exponential increase or decay, and exponentially increasing or decaying dynamics → limited to fixed point and unstable dynamics
- Truly nonlinear systems can be approximated as piecewise linear (sLDS), but interpretability suffers (many tiny states)
- Input-driven dynamics are ignored unless explicitly included in the state space
- Non-uniqueness of the latent space: Many solutions generate the same observations
- Markovian switching dynamics might not be the full picture: longer time scale dependence, hierarchical temporal structure

# Project idea



# References

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